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On the Definition of a Benchmarking Framework for the Fatigue Analysis of Offshore Wind Turbines

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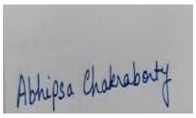

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Abstract

Floating offshore wind energy has received rapidly increasing attention over the past decade. The main concern of the present study is the fatigue reliability analysis of different floating offshore wind turbine technologies, especially their mooring lines. The questions which need to be solved through this research are “How the change in environmental conditions will affect the moorings, platform and turbine characteristics?”, “How the loading on mooring lines changes based on different scales (rated power) of floating offshore wind turbines?”, “How the fatigue analysis can determine the lifetime of moorings”. Presently the novel frontier of offshore wind power is floating wind turbines. There are six vital groups of design: Tension-leg platform, Semi-submersible, Spar, Barge, Combination-type, and Hybrid platforms. The mooring system can be classified into two types: (semi) taut mooring and catenary mooring. The mooring systems contain single-point and multi-point moorings. Synthetic fibre ropes are mostly applied in taut mooring systems. Six strand steel wires contain the broadest application, and multi strand steel wires generally have prolonged service lives. The axial stiffness of spiral strand steel wires is mostly high, and they are relatively used in deep-water mooring systems. Steel wires have reduced corrosion resistance. Modelling floating offshore wind turbines is difficult due to the strong coupling between the hydrodynamics of the floating platform and aerodynamics of the turbine. Normally the mooring lines of the floating wind turbines have six degrees of freedom. OrcaFlex is finite element commercial and time domain software used to model the reaction of cables or to couple trait between a surface vessel and its moorings. Moorings also highly relies on location as it is strongly dominated by water depth and the seabed conditions. In floating offshore wind turbines, sometime misalignment can happen due to the integrated effects of wind movement and wave direction. Simulations are performed on 5MW, 10MW and 15MW based on different environmental conditions and obtained numerous cases. These load cases are then post processed using statistical analysis and then graphs are plotted where statistical parameters are in y axis and the environmental conditions are in x axis to obtain the trendline and R^2 value for determining which environmental condition have the maximum influence on different parameters of moorings, turbine and platform. The fatigue analysis is performed using the rainflow method. The fatigue damage of all the load cases is obtained. Inversing the total damage provides the lifetime of moorings. It is concluded at the end that the lifetime of spar type is more than lifetime of semi-submersible. Among the dynamics and rotation (six degrees of freedom), the surge motion is most affected by environment among the dynamics. The velocity and acceleration are more affected by environment than the effective tension of mooring. Fatigue degradation increases with the increase in scale of wind turbine hence the lifetime of moorings also decreases.

Keywords

design, fatigue, environmental conditions, lifetime, influence, moorings

Contents

Acronyms and Abbreviations:

FOWT: Floating Offshore Wind Turbines.

TLP: Tension Leg Platform.

NREL: National Renewable Energy Laboratory.

DTU: Technical University of Denmark.

DHI: Danish Hydraulic Institute.

SPIC: State Power Investment Corporation.

RWT: Reference Wind Turbine.

OC3: Offshore Code Comparison Collaboration.

KRISO: Korea Research Institute of Ships and Ocean Engineering.

IEA: International Energy Agency.

SEPLA: Suction Embedded Plate Anchors.

DIAS: Design Inspection Analysis Survey.

FAST: Fatigue Aerodynamics Structure and Turbulence.

OC4: Offshore Code Comparison Collaboration Continuation.

FEM: Finite Element Method.

HAWT: Horizontal Axis Wind Turbines.

VAWT: Vertical Axis Wind Turbines.

IEC: Internal Electrotechnical Commission.

RAOs: Response Amplitude Operator.

WAMIT: Wave Analysis MIT (MIT=Massachusetts Institute of Technology) (Software for computing wave loads).

API: Application Programming Interface.

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INTRODUCTION

Floating offshore wind energy has received rapidly increasing attention over the past decade. The difficulty and integrating floating offshore wind turbine (FOWT) systems, as well as the existence of multiple excitation roots, making floating wind upgradation crucial. As such, progressive design and modelling devices sufficient for accurately apprehending these systems' physical performance under practical situations are important. To this end, it is necessary to comprehend and acknowledge the reasons for imprecision and riskiness in these devices in order to evolve an affordable design device for floating wind turbines. Disciplines related to FOWTs have currently encountered considerable research recognition because of the prospective advantages of these systems over onshore ones. Sea currents, waves mooring systems and hydrodynamics, have all been modelled to reproduce the supplementary physical occurrence associated with offshore wind turbines. To construct floating wind turbine technology more cost effective, the power output of independent wind turbines should be elevated, and reliable designs of floating offshore wind turbines (FOWTs) are necessary. In the structural design of FOWTs, both the fatigue-induced failure and the ultimate load-induced failure should be examined appropriately. Moreover, due to the unreliability associated with structural models, load calculations etc., the reliability-based design technique is crucial for the structural security of wind turbines [1]. The main concern of the present study is the fatigue reliability analysis of different floating offshore wind turbine technologies, especially their mooring lines. Dynamic simulations will be performed to determine the loads to which the mooring lines are subjected, and this will later help to analyze the fatigue in the mooring lines and estimate the lifetime of the mooring lines. Analysing the fatigue will help to design more precise and effective mooring lines and dynamic cables which will have a proper balance between the cost and reliability. Designing cost effective and dependable mooring lines and dynamic cables will help the floating offshore wind turbines sector to emerge and in turn will reduce the level of greenhouse gases and carbon emissions. Wind, waves and currents lead to cyclic loading that affects the lifespan. Fatigue failures lead to costly repairs downtime and potential environmental damage. The challenges by the environment may include gusts, wind shear and turbulence, wave induced motion, current-driven vortex induced vibrations [2]. The questions which need to be solved through this research are "How the change in environmental conditions will affect the moorings, platform and turbine characteristics?", "How the loading on mooring lines changes based on different scales (rated power) of floating offshore wind turbines?", "How the fatigue analysis can determine the lifetime of mooring lines". The next chapter deals with the literature review (discussing different research conducted on fatigue analysis). The later chapters deal with methodology, result analysis obtained after simulations, discussion (comparison of how parameters are changing based on scale and platforms) and conclusion.

1. FLOATING OFFSHORE WIND TURBINE TECHNOLOGY

1.1 FLOATING OFFSHORE WIND TURBINE TECHNOLOGY PLATFORM TOPOLOGY

Presently the novel frontier of offshore wind power is floating wind turbines. Its due to the possibility of their installation in deep water where wind speeds are always higher. The novel designs have floating turbines, that bob and sway with the waves and wind, anchored with chains or fixed with ballast to the seafloor. There are six vital groups of design: Tension-leg platform (TLP), Semi-submersible, Spar, Barge, Combination-type, and Hybrid platforms [2]. In Fig 1 and Table 1 the overview of different configuration is provided.

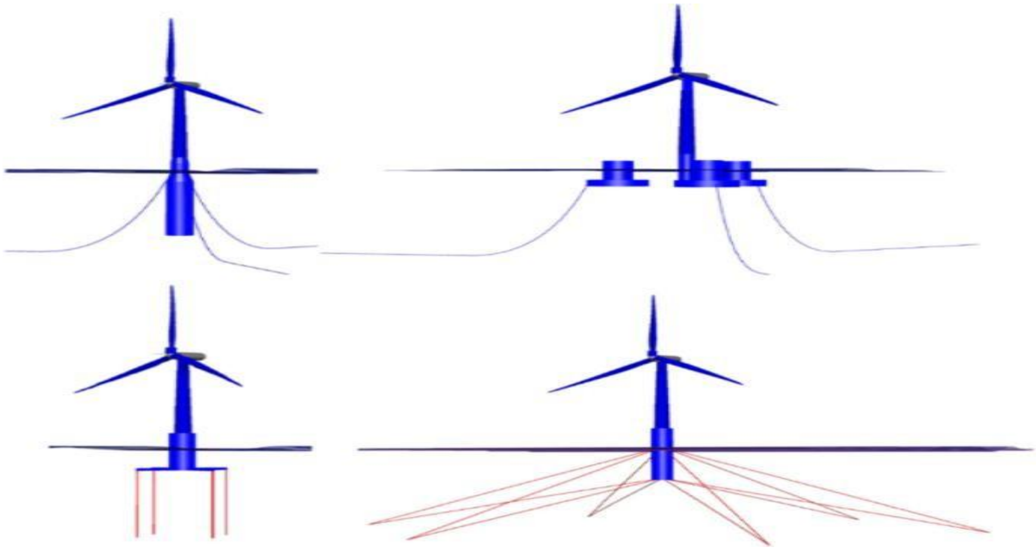


Figure 1: *Offshore floating concepts, clockwise from top-left: Spar, Semi-Submersible, Tension Leg Buoy, and Tension Leg Platform. [3]*

Table 1: *Difference between various floating offshore wind turbine platforms.[5]*

Table 2. Comparison of floating platforms for FWTs.

Platform types / Aspects	Semisubmersible	Spar	TLP
Popularity	Most popular concept	Has been popular	Not widely deployed
Water depths	Main ≥50m	Deeper water (≥120m)	Main ≥50m
Operational risks	Low risk (higher exposure to waves, potential problems for cables)	Low operational risk (high fatigue loads in tower base)	Higher operational risk (high vertical load in the moorings and anchors)
Hydrostability	Less stable (distributed buoyancy stabilized, requires moveable water ballast to limit tilt)	Inherent stability (ballast stabilized, achieves stability through ballast in buoyancy tank)	High stability, low motions (mooring line stabilized, static stability through mooring line tension)
Mooring types	Catenary mooring, easy installation	Catenary mooring, easy installation	Vertical mooring, complex installation
Suitable anchors	Mainly suction anchor, fluke anchor, anchor pile	Mainly suction anchor, fluke anchor, anchor pile	Mainly suction anchor, anchor pile
Soil conditions	Insensitivity	Insensitivity	Influenced by the selected anchors
Weights	Weight for 6MW: ~3000t	Weight for 6MW: ~3500t	Weight for 6MW: ~2000 t
Costs	More economical	High cost, 5–8 mEUR/MW	The most expensive due to complex mooring systems
Fabrications	Large and complex structure, complicated in fabrication	Complex manufacturing	Lower material costs owing to smaller structural weight
Transportation and installation	Simple (requiring the specialized vessel)	Complex (requiring specialized installation vessels)	Difficult (unstable during assembly, requiring special vessels)

Barges are shallow, wide platforms that utilizes buoyancy remote from the centre of the structure to restrain the wind force on the tower. As they normally stretchless more than 10 metres underwater, they do not require any advanced deep-water docks or installation vessels. However, they can be challenging to construct because the platform is normally a single, large unit with a complicated shape [3]. In Fig 2 a schematic structure of barge is described.

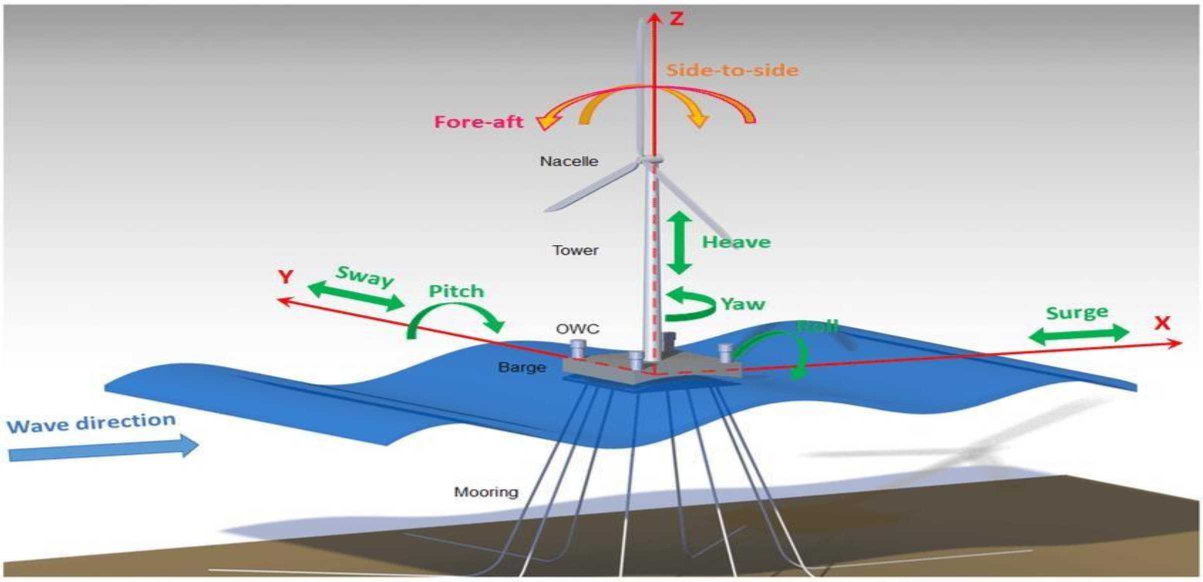


Figure 2: *A clear structure of a barge type Floating Offshore Wind Turbine.[4]*

The hybrid platforms introduce another category of renewable energy, mostly called a wave energy converter as seen in Fig 3. They combine offshore wind turbines with wave energy on a shared platform A wave energy converter also lowers platform motion, which in turn maximizes the power performance from the turbine [3].

Figure 3: *PelaFlex platform with wave energy converters and wind turbines.[6]*

1.2 FLOATING OFFSHORE WIND TURBINE: TECHNOLOGY & SCALES (EXAMPLES OF SOME PREVIOUSLY CONDUCTED EXPERIMENTS)

A study based on scaled prototype tests for the National Renewable Energy Laboratory (NREL) 5 MW wind turbine assisted on three types of floaters, mostly a tension leg, a spar-type floater as shown in Fig 4, and a semi-submersible was executed. A comparison investigation was held among the dynamic reaction of these three models under wind-wave conditions, which demonstrated that the spar-type floater describes the slightest surge reaction, the tension leg owns the slightest pitch response, and the semi-submersible exhibits the most important 2nd-order low-frequency response related to surge and pitch motions under non-uniform waves. In fact, the surge response of the spar-type floater is also quite reliable on the mooring system [28].

The 1:60 scale Technical University of Denmark (DTU) 10 MW turbine was examined on a TLP floater with three control plans. The TLP floater was developed by Korea Institute of Energy Research (KIER) and the tests executed in the deep-water basin of Danish Hydraulic Institute (DHI) Denmark. The experiments established that application of onshore controller settings guides to stronger blade pitch action and surge motion than for a de-tuned controller, where the response time was longer. The motions were even smaller, when running with blade pitch and fixed rotor speed [21].

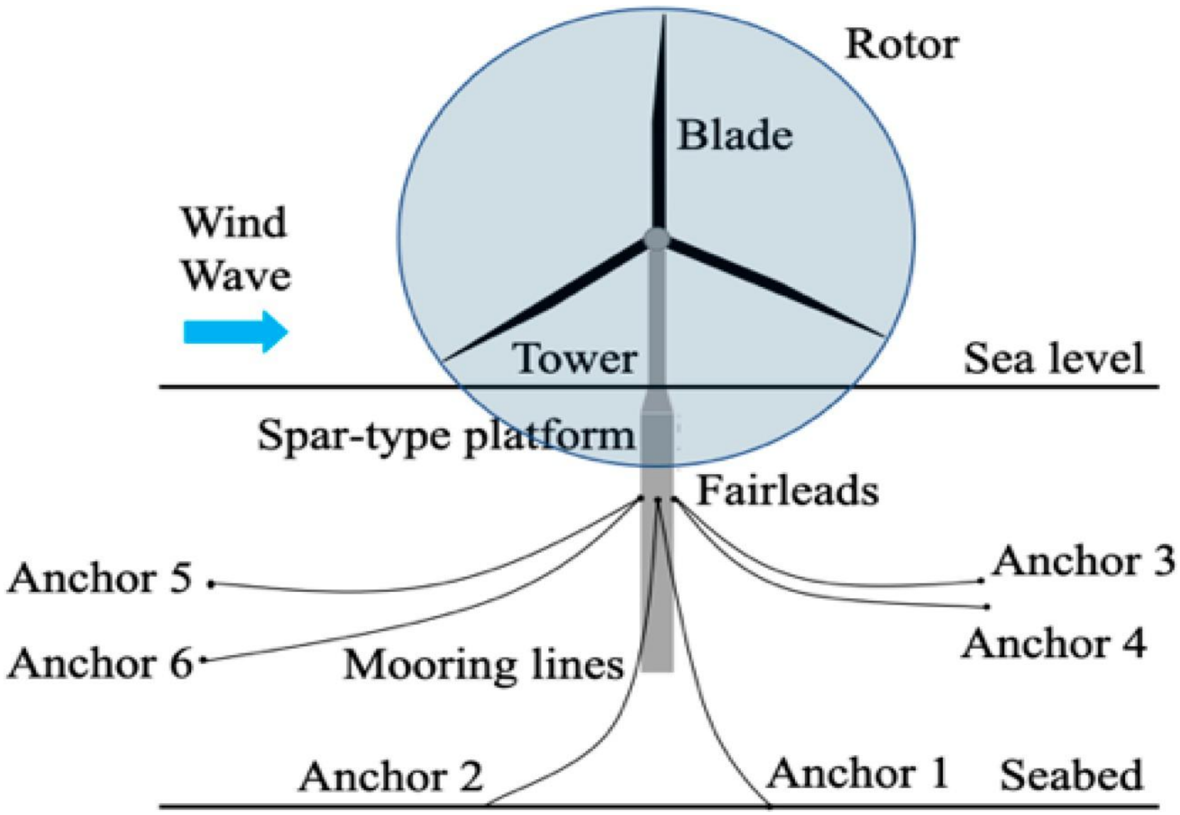


Figure 4: *5MW spar type floating wind turbine.[28]*

A sequence of numerical simulations and experiments were carried out to examine the dynamic responses of the 10 MW State Power Investment Corporation (SPIC) wind turbine system at halfway water depth. The experimental models and infrastructures were initiated first to secure the perfection of the basin model test. Then, a numerical model was developed based on the fully coupled aero-hydro-servo-elastic-moor technique. Operational non-uniform waves, steady winds at rated wind speed, and turbulent winds with variable turbulence intensities (TIs) were chosen as the environmental conditions. The numerical outcomes of the horizontal restoring stiffness, free decay analysis, white noise analysis, wave-only tests, wind-only tests, and wind–wave integrated tests were compared with the experimental outcomes, respectively, to confirm the rationality of the numerical model [26].

The spar platform used in this 10MW (Reference Wind Turbine) RWT was developed according to the outline for the Offshore Code Comparison Collaboration (OC3) Hywind spar platform, which is described by Jonkman. The draft of the hull for this 10 MW platform is decreased to 90 m in contrast to the 120 m in the OC3 Hywind design, which makes it more acceptable to be installed in halfway water depths where the costs of installation and transportation are lower than in deep water. Meanwhile, the hull diameter is raised to provide an adequate buoyancy. Heavy ballast is positioned at the bottom of the hull to balance the platform. A mooring system consisting of clump weights and three catenary lines is applied in this model. As in the OC3 model, a rotational spring is attached to attain an adequate yaw stiffness of the mooring system [25].

A 15 MW multi-column TLP-type wind turbine platform (KRISO-TLP) moored with the wire rope tendons was developed for the east offshore of Korea. Design characteristics of the KRISO-TLP are given as: (1) a shallow lightship draft with the turbine permits tow-out activities from shipyards in Korea, (2) the platform has a self-stability for wet-tow and installation, and (3) square-shape columns can promote hull fabrication at small shipyards. This perspective may require some hull weight penalties in contrast to the mono-column TLP but can eliminate all the risks, costs, and complicated functions connected with preservice execution. The aim of the study is the principal design of the KRISO-TLP to support a 15 MW turbine, reshaped from an NREL/IEA (International Energy Agency) 15 MW turbine offshore application. The original turbine designed for the semi platform was remodelled for the current TLP application [22].

Dynamics of the VolturnUS-S (floating concrete structure that support wind turbine) semi-submersible platform designed for the IEA 15 MW wind turbine under freak waves is investigated using an integrated computational fluid dynamics and finite element mooring line prototype. First, the hydrodynamic model is facilitated using the DeepCwind(US based project) semi-submersible platform. Then, the facilitated model is employed for the analysis of the VolturnUS-S semi-submersible platform under freak waves demonstrated as a concentrated group of waves. Hydrodynamic characteristic of the semi-submersible FOWT is analysed in detail, and parametric analysis of the problem containing the effects of focusing location, gravity centre location focusing crest amplitude, incident angle, is conducted [23].

An investigation on the structural modelling, design, control system, and considerations of 20 MW spar FWT (Floatting Wind Turbine) had taken place, and an extensive dynamic and structural analysis was performed, concentrating on the relative significance of wind and wave loads on the fatigue life and utmost stresses at varying locations of the platform and tower. Three spar FWTs supporting the 20 MW wind turbine from Ashuri et al. are achieved from a parametric design procedure, with the diameters at the bottom and at sea water level (SWL) as design variables. Hydrodynamic loads are assessed with potential theory and divided over the hull. A controller with motion compensation based on feedforward of the nacelle velocity is acquired, ignoring the instabilities associated with controller-motion interactions. The analyses are based on fully integrated, non-linear time-domain simulations, using an aero-hydro-servo-elastic software [27].

Table 2: *Wind Turbine information for different scales of Floating Offshore Wind Turbines.*

Rated Power	5MW	10MW	15MW	20MW
Rotor Diameter	126m	178.3m	240m	276m
Hub Height	90m	119m	150m	160m
Cut in wins speed	3 m/sec	4m/sec	3m/sec	3m/sec
Cut out wind speed	25m/sec	25m/sec	25m/sec	25m/sec
Tip Speed	8.3rpm	9.6rpm	7.6rpm	8.3rpm
Rated Speed	11.4m/sec	11.4m/sec	10.56m/sec	10.7m/sec

2. OVERVIEW OF MOORING SYSTEM AND DYNAMIC CABLES

2.1 MOORING SYSTEM TOPOLOGIES

According to mooring line layout as shown in Fig 5 the mooring system can be classified into two types: (semi) taut mooring and catenary mooring. Catenary mooring strategies are mainly embraced in shallow water, and the line static layout can be explained by the catenary equations. The mooring line gravity supplies the restoring forces. Taut mooring is usually applicable to deep water. The taut line has a definite inclined angle with the seabed, and the huge tension is conserved in the line. Taut mooring systems mainly utilizes superficial mooring lines, e.g., synthetic fibre ropes. [12] However, anchors for the taut mooring are required to withstand both vertical and horizontal leads, which are ambitious in installation and design in deep sea. For TLPs, there is a specific mooring system consists of multiple vertical tendons/tethers which are assembled of steel tubes/fibre composites. The vertical mooring system permits horizontal movements under wave loads but demoralize much heave, roll, and pitch motions, which makes the TLP a favoured alternative for stability [8].

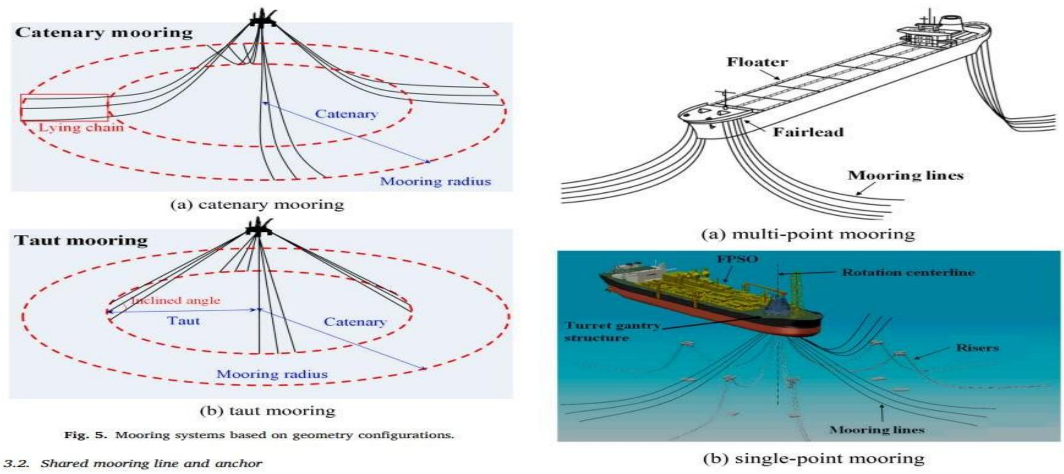


Figure 5: *Different types of mooring lines.[8]*

Based on the interconnected points at the fairleads, the mooring systems contains single-point and multi-point moorings. The multi-point mooring has dissimilar attached points (fairleads) to join mooring lines. It has broad applications in all types of platforms, basically in permanent mooring systems. Single-point moorings embrace layout with all mooring lines linked at one-point, which mostly needs rotating devices. The environment load can be decreased by utilizing the weather-vane effect, and thus single point mooring is acceptable for inhomogeneous loads. Single-point moorings are usually employed in offloading and floating production storage [6]. A framework of mooring design is shown in the Fig 6.

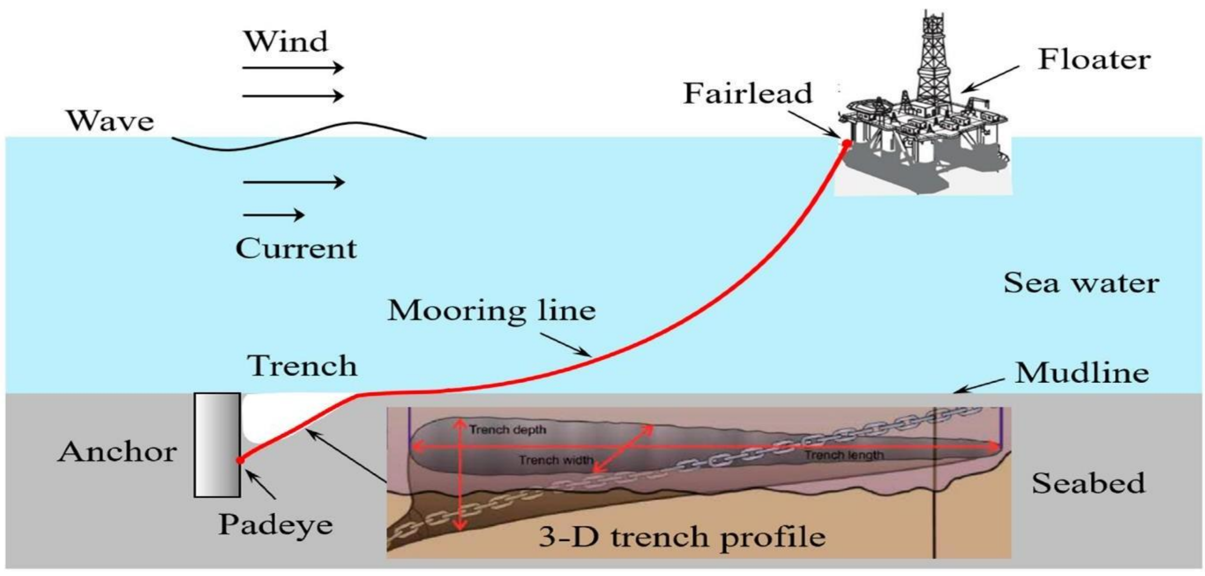


Figure 6: *A Framework for Mooring and Anchor Design. [9]*

Design concepts of mooring systems mostly comprise of two aspects. The first one is to supply restoring forces for the slow-varying movements, which are caused by the wind loads (including mean drift) and second-order wave. The second one is to supply compliance for wave frequency propagation of floating structures. Mooring systems should not hinder the wave-frequency motions of floaters, else very expensive and heavy mooring design schemes will be required to undergo the first-order wave loads on floaters [11]. A superior catenary mooring system design is secured by scheming the nature periods in yaw, sway and surge motions away from the primary wave periods. Therefore, a mooring system is usually designed examining the induced loads due to the direct hydrodynamic loads and the floater motions on the lines. Such design ideas are globally valid and govern the mooring systems for distinct floaters [9].

In the mooring system inspection, the universal hydrodynamic calculations (frequency domain analysis, time domain analysis and coupled analysis) for floaters are accompanied first, mainly using a frequency-domain predictions based on first- and second-order potential flow theory, in which the mooring systems are assumed in an uncoupled (only restoring) way. Then, global motions of floaters with their mooring systems are estimated in a coupled investigation to acquire the dynamic variation of mooring line tension. In an uncoupled mooring investigation, the floater movements are mostly estimated based on the diffraction/radiation hypothesis in which loads from mooring lines are only modelled as restoring forces. Mean loads on mooring lines due to the current can be assumed. In a fully coupled mooring analysis, the non-linear performance of mooring lines is considered in floater motions [22]. The coupling outcome contains dynamic behavior from the damping forces, inertia forces and restoring forces of mooring lines, linear/nonlinear springs. For example, the quasi-static catenary layout can be demonstrated by the catenary equations when only assuming the mooring line gravity in water [12]. In catenary moorings, the loads transmitted by mooring lines are horizontal loads. Gravity anchors can be acquired by deploying the friction resistance between the seabed and anchor. Suction and piles anchors can withstand large horizontal capacity with the enhanced load position about 0.65–0.75 times the implanted depth. Fluke anchors are also acceptable to extract horizontal loads, and the penetration depth increases when the loads exceed the previous maximum loads. Suction embedded plate anchors may be adopted in the catenary moorings, but the plate inclined angles need to be carefully managed during the 'keying' process. Torpedo anchors have eyes on the tip and show visible rotation under the horizontal loads [17]. Screw anchors are inappropriate for horizontal loads because of the plate directions. Anchors for the TLPs are only exposed to vertical loads. Gravity anchors can be used by the immersed weights. In clay, uplift resistances of suction and piles anchors come from the outer friction resistance and reverse bearing capacity in undrained conditions. In the sand, outer and inner friction resistances provide the pile resistance in drained conditions. For suction anchors, the suction resistance may be deployed depending on the sand permeability and loading rate. Screw anchors are acceptable for vertical loads because of the increased plate area to mobilize the vertical resistances. Other kinds of anchors, e.g. Suction Embedded Plate Anchors (SEPLAs), Design Inspection Analysis Survey (DIAs) and fluke anchors, are seldom used to resist vertical loads. When the inclined angle is lower than 15°, the anchor behaviour is predicted by the lateral resistance. When the inclined angle is higher than 45°, the interface strength reigns [20]. Different parts of the moorings are depicted in Fig 7 and Fig 8.

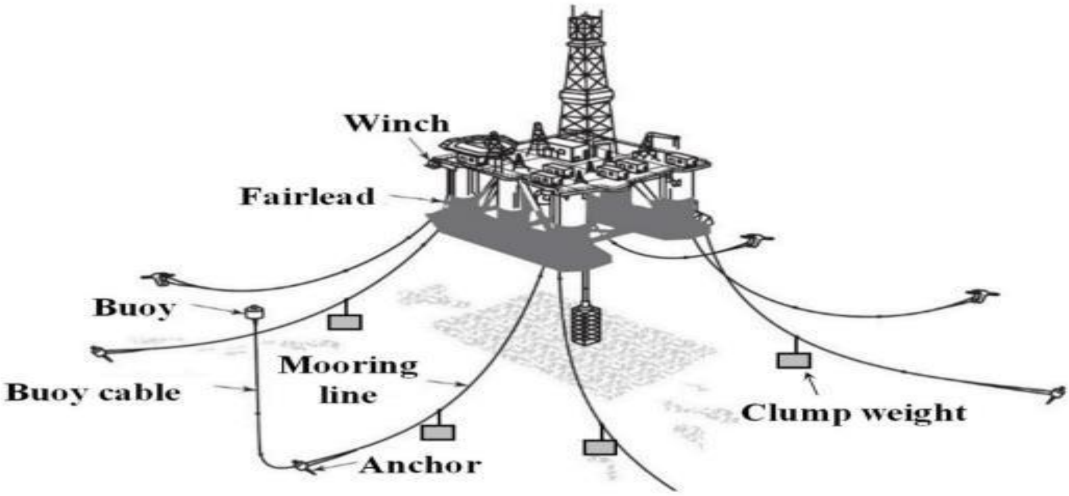


Figure 7*: Mooring system showing different parts of it.[11]*

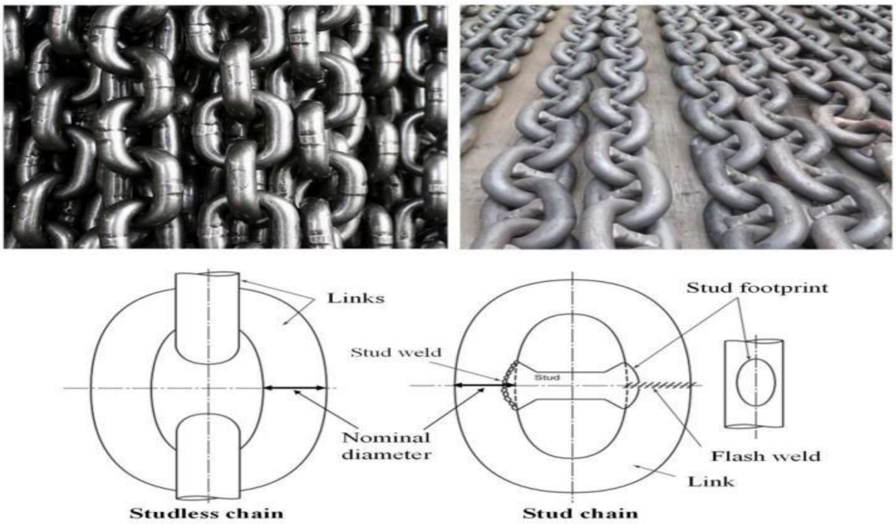


Figure 8: *Types of mooring chains.[6]*

2.2 MOORING SYSTEM MATERIALS

Mooring chains, comprising of chain links attached one by one, are the utmost widely used component. The chains are acceptable to catenary moorings because of its large weight, wear resistance high strength, etc. Mooring chains can be split into stud less chain and stud chain depending on their structures. Mooring chains also have some limitations in some applications. First, its manufacturing cost is relatively higher compared with that of the steel wire and fibre rope. Second, it is infrequently used as a single component in deep water because of its large weight. Third, stud less chain may encounter knots in temporary mooring [5]. Synthetic fibre ropes are mostly applied in taut mooring systems, and are featured by lightweight, high strength, etc. presents numerous types of synthetic fibre ropes, inclusive of polyester fibre cables, nylon cables, polyethylene cables, etc. Among them, polyester fibre materials have good fatigue performance, strong creep resistance, high strength and low cost, which can be engaged in long-term systems [6].

Steel wire is comprised of multiple strands of steel wire perverted around the core, presents regular structures of steel wire, consisting of six strands, eight strand, multi strand, and spiral strand. Six strand steel wires contain the broadest application, and multi strand steel wires generally have prolonged service lives. The axial stiffness of spiral strand steel wires is mostly high, and they are relatively used in deep-water mooring systems. Steel wires have reduced corrosion resistance [17].

2.3 DYNAMIC CABLES TOPOLOGIES

Power cables are broadly utilized in the electrification of floating oil and gas generation infrastructures, and power transmission lines where they have to tolerate significant cyclic loads induced by the union of floating body dynamics with wind, waves and current impacts. One of the crucial design problems for power cables in marine applications is the fatigue strength; fatigue assessment studies have described that errorless analysis of the complicated dynamic behavior induced by the offshore environment should be carried out for fatigue assessment purposes. A comparatively new application field for marine power cable constituents is floating offshore wind turbines (FOWTs)and farms. FOWTs, along with their mooring systems and linked power cables, must tolerate numerous environmental activities, such as correlated variance in the wave, current, and wind loads, which cause uninterrupted and variable dynamic motion and stress in the power cables that attach the single FOWTs with each other and/or with the land. The collection of variable stress results in cumulative injury to the fatigue. The framework of a dynamic power cable as shown in Fig 9 mostly consists of several material ordered in numerous layers integrated in a cylindrical or helical layout. The power cable can be assumed as a multilayer, non-bonded composite flexible structure. Such structures demonstrate non-linear mechanical behaviour resulting from friction and sliding between constituents. Due to torsional stiffness and low bending features, dynamic power cables are malleable in bending and torsional degrees of freedom while having a huge axial stiffness [7].

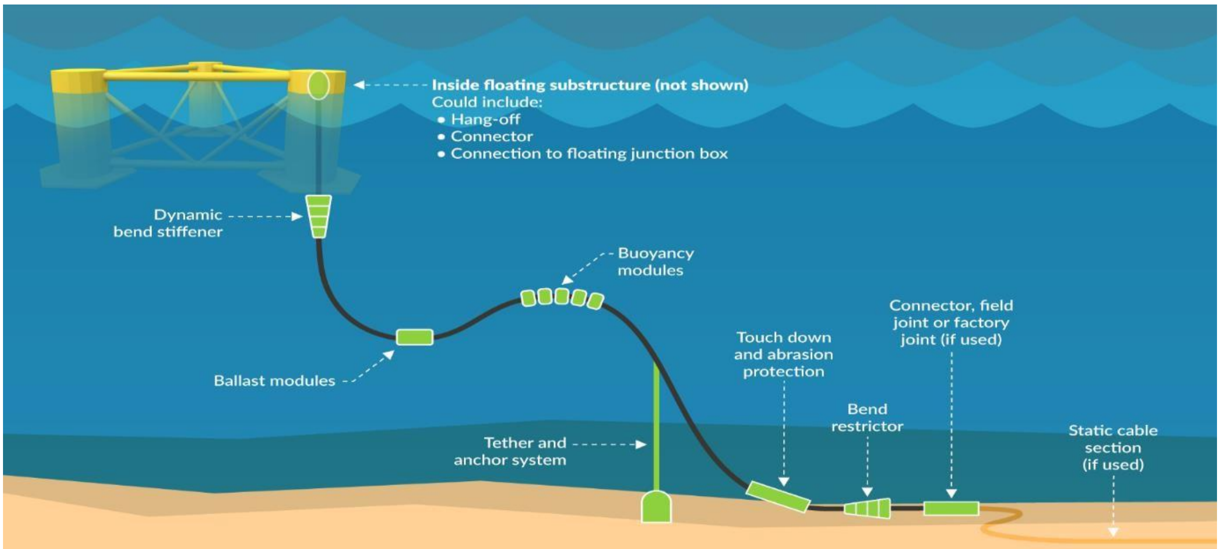


Figure 9: *The dynamic cable system of floating offshore wind turbines.[7]*

In offshore wind farms for floating wind turbines, umbilicals and power cables are broadly utilized to control the transmit power and system from the offshore to grid, and they are subjected to a complicated fluid–cable–soil interaction system after they are equipped on seabed. One typical difference between the offshore wind farm and the onshore wind farm is the usage of dynamic umbilicals and cables. The on-bottom steadiness of umbilicals and cables is a design challenge for the offshore wind industry and most of the designs are very conventional. The distinctive properties of umbilicals and power cables compared to conventional pipelines are lightweight, small in diameter, and have a low bending stiffness. Due to these properties, umbilicals and power cables are subjected to experience imprudent lateral displacement, and then the on-bottom steadiness becomes more difficult [10]. The excessive lateral displacement will cause the injury of umbilicals and power cables, clash with each other and the shutdown of wind farms. The dependable design of the on-bottom firmness of umbilicals and power cables for offshore wind application is important for the integrity and safety of the wind farms. The electrical cables that link the FOWTs in floating offshore wind farms are of the umbilical typology, supplying a service-support from the original station to the single FOWTs. Such umbilical cables belong to the optical powered submarine type and are provided to contribute electrical power or to convey operating control inputs and data to underwater constituents in offshore oil and gas infrastructures. Umbilical cables are executed in floating offshore wind farms and linked to each FOWT for the same requirement; therefore, their service ability and integrity are crucial for the correct operation and coordination of the FOWTs within the farm. The stress situation in such non-homogeneous cross-sections of the umbilical cables is quite complicated, with non-linear local stress due to friction and contact between the components of the tube, and the aleatory and model unpredictability due to this complication is vital for the reliability of any fatigue analysis [12]. Different types of FOWT connections are depicted in Fig 10.

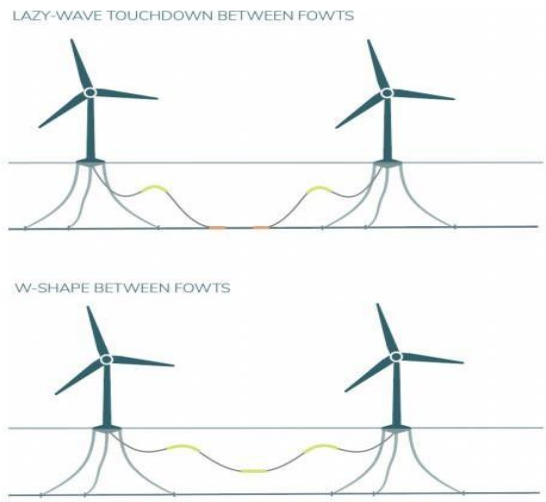


Figure 10: *Types of connections between floating offshore wind turbines, Lazy wave, W-shape.[10]*

2.4 DYNAMIC CABLE MATERIALS

Table 3: *Overview of the insulation materials in dynamic cables.[12]*

Material		Use	Considerations
XLPE	Cross Linked Polyethylene	Modern Array Cables	Modern co-polymers allow XLPE systems to be used in all applications for standard radial field power transmission cables. Moderate bend stiffness. Significantly Lighter than ERP.
EPR	Ethylene propylene rubber	Some Array Cables	Significantly heavier than XLPE. Dielectric loss prohibits use for HV at reasonable stress. Low bending stiffness. Typically, more Expensive. Lower hot viscosity so has a greater tolerance to contamination. Easier to joint.
Oil/Paper	Impregnated Paper	Older Systems Land Cables	Lower operating temperature. Can be expensive to manufacture. Due to the oil impregnation process manufacturable length is restricted in comparison to XLPE and EPR. Termination requires sealing ends carefully and can be harder to achieve. With a lead layer they can be difficult to install and have poor fatigue resistance.

For component design the size, stiffness, weight, strength, design life, fatigue resistance function, minimum bend radius, length, cost and water penetration assumptions are evaluated. If long lengths are needed jointing of cable components is also a significant factor and the quantity and number of jointing can affect component performance. Component joints evolved for the static cable windfarm application are planned only for decreased level fatigue and tension levels and are thus never present within any sections which may be exposed to continual loading or places under significant tension (e.g. in the portion of cable from the seabed floor to the elevated termination point within the wind turbine structure during installation) [13]. Choice of material for conductors have a direct influence core size. Conductors for windfarm applications are usually either copper or aluminium. Increased conductivity means copper conductors are small. Decreased conductivity means aluminium conductors require to be larger than copper conductors to meet current carrying requirements to transfer the same amount of power. Since 2004, copper has become comparatively more costly than aluminium. Essential investment in preliminary aluminium production was allured as replacement of aluminium for copper became surprisingly popular in the aerospace and automobile industries, alongside China’s accelerated growth demands, resulting in a surplus on the global markets. In comparison, generation has remained relatively stable while demand has enlarged for copper resulting is price increase. Copper cores have higher fatigue resistance being able to tolerate larger vibration amplitudes over prolonged durations than aluminium without breaking or cracking. Copper exhibits low levels of creep in contrast to aluminium and is also less subjected to failure due to the respective oxide characteristics; copper oxide is conductive, soft and breaks down easily whereas aluminium is electrically insulating, which can also make jointing more challenging [10]. There are some differences between using copper and aluminium as a dynamic cable material in Table 4.

Table 4: *Overview of the conductor materials in dynamic cables.[12]*

Copper	Aluminium
• High Conductivity • High Density (Heavier) • Higher Cost Material • Widely available • Easy to Process	• Lighter • Lower Conductivity (≈60% compared to copper) • Lower Tensile Strength • Fluctuation in price w.r.t. Copper • Highly reactive metal

3. NUMERICAL METHODOLOGY

Modelling FOWT is difficult due to the strong coupling between the hydrodynamics of the floating platform and aerodynamics of the turbine. A wide range of software packages and numerical methods are available for the predominant stages of the design procedure of a FOWT prototype. Some overarching aspects and contemporary directions are assumed in this section. Traditionally, a numerical model of a novel design is created, and then verified using model-scale laboratory tests. However, in the ride for cost reduction of FOWT technology designers may be focussing to use high-fidelity numerical tools in an attempt to decrease reliance on expensive and time-consuming physical tests, and also to lower the unpredictability that simpler numerical models contain [16].

3.1 MAIN DYNAMICS AND LOADS

Dynamic motion properties of floaters, comprising natural periods, is one vital characteristic that should be assumed in the design. The primary design concept is to escape the resonant motions resulting from first-order wave loads by having natural periods higher than the important wave periods (5–20 s) or smaller than those periods for TLPs. Normally the mooring lines of the floating wind turbines have six degrees of freedom as given in Table 5 [16].

Table 5: *The six degrees of freedom and wave periods.[17]*

Types / Modes	Semi-submersible	TLP	Spar	Ship-type
Surge	>100s	>100s	>100s	>100s
Sway	>100s	>100s	>100s	>100s
Heave	20–50s	<5s	20–35s	5–12s
Roll	30–60s	<5s	50–90s	5–30s
Pitch	30–60s	<5s	50–90s	5–12s
Yaw	>100s	>100s	>100s	>100s

Integrated vertical and horizontal loads are commonly encountered in practice, mostly in the semi-taut mooring. Thus, the coupling impacts of vertical and horizontal loads should be considered. One technique to detect the combined capacity is to adopt the failure envelopes under different load directions. Another technique is to use plastic limit analysis to calculate the ultimate anchor capacity [14].

It is essential to elucidate the load history accomplished by dynamic cables, including external environmental elements (such as wind, waves, and currents) and the loads originated by the cable's vibration. Since dynamic cables are malleable cables with high damping properties and are less subjected to vortex-induced vibrations, the effect of vortex-induced vibration on the fatigue injury of dynamic cables can be ignored in the line design. Dynamic cables are even exposed to harmonic, tensile and bending loads [21].

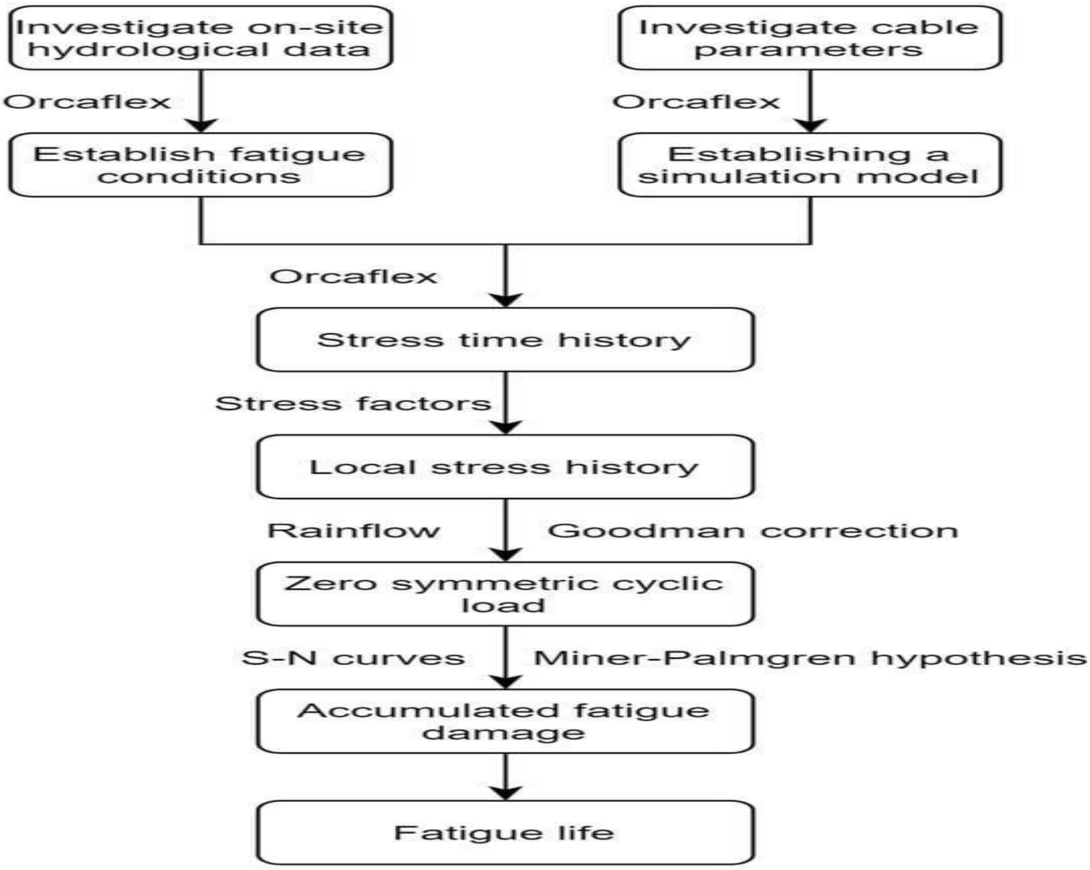


Figure 11: *OrcaFlex predicting time domain fatigue analysis of wind turbines.[18]*

3.2 SIMULATION SOFTWARES

OrcaFlex is finite element commercial and time domain software used to model the reaction of cables or to couple trait between a surface vessel and its moorings. Such models are mostly recognized as independent cable models because the mooring line is optimized as a system of mass elements (nodes) linked to visco-elastic components. The stiffness, damping and inherent mass features of the mooring confirms it reacts to numerous displacements and end forces. This results in electrified loads that cannot be predicted by the quasi-static model. OrcaFlex is mainly used in the offshore industry as a way to analyse the mooring line snap loads, swift re-tensioning, and transient motion. As an additional advantage, OrcaFlex also supplies a graphical user interface (GUI), but FAST does not; therefore, OrcaFlex is convenient in the visualization procedure. OrcaFlex is designed to be vigorous for a variety of cable layout. OrcaFlex assists both time-domain and frequency-domain dynamic analysis. The frequency-domain analysis is a linear process. The frequency-domain solver imprecise any nonlinearities through the procedure of linearization. The nonlinearity of time-domain analysis is more extensive since at each mass, damping, stiffness, loading, time step and other attributes are assessed, taking into assumption the instantaneous, time-varying geometry. OrcaFlex uses a 3-D finite element model to simulate mooring line dynamics. To model the bending, axial and tortional stiffness and damping of lines, they are independent as lumped-mass components linked to visco-elastic spring-damper segment [11].

FASTlink is an integrating module being created by Orcina in association with industrial partners to utilize the robustness of FAST (fatigue, aerodynamics, structures and turbulence) and OrcaFlex to construct a high-fidelity modeling equipment for offshore wind turbines. The wind turbine, its control system, aerodynamic loads, tower and the six degree-of-freedom rigid-body platform motion is modeled in FAST. The subsea elements, such as the support platform hydrodynamics and mooring lines, are modeled in OrcaFlex. Therefore, when FASTlink is being used, HydroDyn is no longer performing calculations. Generally, as part of this coupling procedure, the quasi-static cable model in FAST is impaired. Mooring forces would then be demonstrated in OrcaFlex. In the FASTlink coupling module, FAST drives the vessel position and velocity vectors into OrcaFlex; the hydrodynamic added mass matrix non-acceleration dependent hydrodynamic forces and line tension are calculated by OrcaFlex; the final added mass matrix and total force and moment on the platform is transferred back into FAST; then the ultimate platform motion is answered in FAST [24].

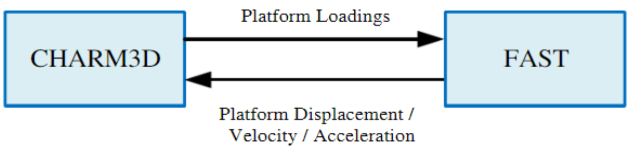


Figure 12: *Basic concept of CHARM3D-FAST coupling.[24]*

An in-house time-domain completely coupled dynamic simulation program “Charm3D-FAST as shown in Fig 12 is utilized for the simulation of the OC4 semisubmersible wind turbine. The wave forces on the structure under uneven wave conditions are estimated using the first and second-order diffraction/radiation potential theory. The asymmetric viscous drag forces were estimated from Morison’s formula at the instantaneous body position and up to the instantaneous free-surface advancement. The modelling of the mooring system dynamics is based on a higher-order rod Finite Element Method (FEM) [25]. FAST is an open-source software originated as a research equipment by the National Renewable Energy Laboratory (NREL) in the United States. Currently, it was re-named OpenFAST, with the concept of having a more community-driven growth of the equipment. OpenFAST is made up of multiple modules able to answer the coupled nonlinear aero-hydro-servo-elastic-mooring dynamics of FOWTs in time-domain only. Although several scholars have managed comprehensive research on the mooring systems of offshore floating wind turbines, there needs to be more experimentation on the impacts of different line materials on the dynamic response of semi-submersible wind turbines using OpenFAST [23]. This requirement for experiment may be because of the restricted usage of fiber ropes in floating wind turbine mooring systems. FAST contains a number of modules that utilizes mathematical models to simulate one or more turbine elements, such as HydroDyn (hydrodynamics of platforms for offshore structures), ServoDyn (control and electrical systems), AeroDyn (aerodynamics), and BeamDyn or ElastoDyn (structural dynamics). The time-domain simulation and examination of coupled nonlinear aerohydroservoelastic simulation of onshore or offshore wind turbines on the floating substructure or fixed-bottom is made possible by the gathering of these modules [18].

QBlade is an excellent multi-physics software developed to assist the certification, aero-servo-hydro-elastic development and prototyping, simulation of wind turbines. Qblade has the following roles.

- Wind Turbine Composition (HAWT (Horizontal Axis Wind Turbines), VAWT (Vertical Axis Wind Turbine) and Multi-Rotor)
- Numerical and Floater Design Testing
- Optimization and Controller Testing
- IEC (Internal Electrotechnical Commission) Design Load Certification
- Wake Interaction Studies and Wind Farm Simulations
- Digital Twins in Matlab and Python Environments

OrcaFlex is used in the project because it helps in comprehensive dynamic analysis, it is highly accurate, it has a user-friendly interface and employees advanced numerical methods. It also has some disadvantages as it is mostly dependant on input data, has limited flexibility and have high computational requirements [15].

3.3 STATISTICAL ANALYSIS

There is a huge number of files that are created by the automation by API method, so post processing is needed to compile all the results obtained from mooring lines, turbine and buoy. Another python script is created there the excel file containing the results are first read then different statistical tools are added to calculate the average, minimum, maximum, standard deviation, root mean square, range and 95th percentile of the results. After calculating the statistical parameters, they are again stored in a separate excel file. The graphs of the statistical data are plotted against wave height, wind speed, wave period to analyze which environmental condition has the most and the least influence on the mooring line, buoy and turbine (the value of R square mostly indicates the influence, more the R^2 value more the parameter is sensitive to environmental conditions).

3.4 FATIGUE ANALYSIS

A python script is created where the rainflow package is imported. The SN curve parameters are imported from DNV. The total damage is calculated. The mooring characteristics are included. The estimation of damage is conducted where all the excel sheets created before (during automation by API) are read and the effective tension is recognized for determining the fatigue of mooring line. Finally, the total damage of the load cases is calculated whose inverse determines the lifetime of the moorings [34]. Rainflow is considered as the cycle counting method used in fatigue analysis to identify and count number of cycles in a loading time history. It has several advantages like accurate cycle counting, handles complex loading and provides conservative estimates of fatigue damage. It is mostly applicable to non-periodic irregular loading conditions usually encountered in real-world applications. The methodology involves counting and identifying the turning points (peaks and valleys) in a strain or stress signal and then assembling these turning points into ranges. These ranges, or cycles, are then utilized to build a fatigue damage histogram, supplying insights into the distribution of loadings and aiding in the determination of fatigue life. The major function of rain-flow counting technique is that the estimated load history data which has been eliminating the detection value of peak valley and the invalid amplitude is indicated in the form of discrete load cycle. Since this counting technique could be managed on computer platform, it could be automated and programmed without many complications. In contrast to other counting techniques, rain-fall counting is more precise, so it is highly applied in the research and application of fatigue-related issues. The most fundamental programming process mainly includes the following steps: 1. Data Compression, 2. Extraction of the Number of Cycles [30]. The number of cycles to failure at tension range computed as follows,

$$N = C/((bin[0] ** k) \tag{1}$$

where $d = n/N * 3600 * 24 * 365.25/t[-1]$ is the damage and n the number of cycles within tension range interval. Here C is a matrix that stores the cycle counts for each stress range, bin is an array that defines the stress range bins, k is a variable that represents stress range multiplier and t is the array that stores the time or cycle count values corresponding to each stress ranges.

4. ENVIRONMENTAL CONDITIONS

Hydrodynamic testing of offshore floating platforms can be applied to secure the motion response of the floating platform under the integrated interaction of waves, currents and winds. The direction of current wind and wave with respect to wind turbine can be seen in Fig 13. The experimentation on the dynamic response of a floating platform could be used to determine the platform's wave conditions, wave resistance on the deck, slamming load forecasting, etc. These are based on global finite element analysis and the global strength analysis along with the fatigue analysis of the floating platform. Theoretical analysis, numerical simulation methods and model experimental test are the main equipment to understand the behavior of the platform motion response of marine structures under the activities of wave currents, etc [14].

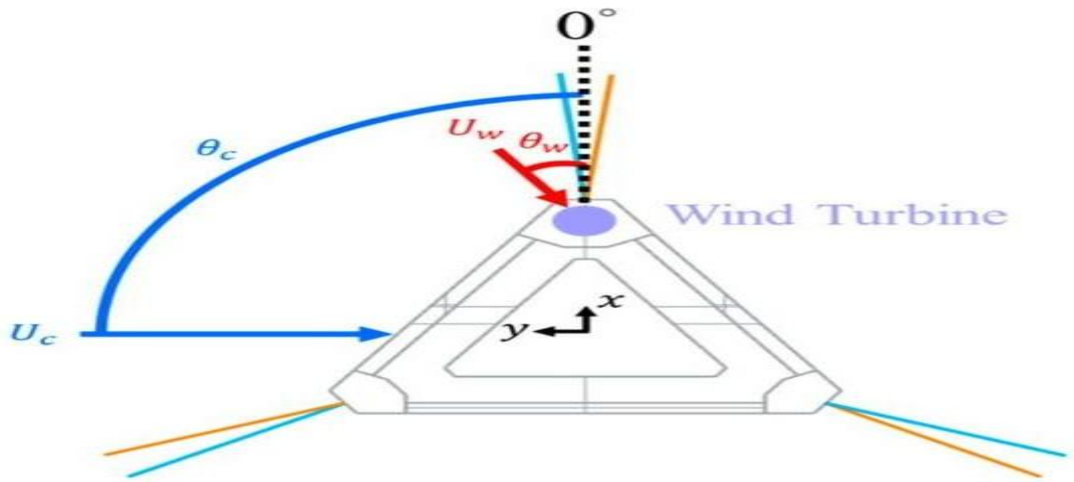


Figure 13: *The direction of current wave and wind towards a wind turbine.[14]*

4.1 WIND, WAVE AND CURRENT CONDITIONS

The marine environment model consists of:

i. Current: Can have a library of current profiles, each with magnitude and rotation varying with depth and/or time.
ii. Waves: Wave models accessible contain a simple linear (sinusoidal) wave, multiple non-linear uniform waves and a number of wave spectra for modelling irregular seas. Random seas can also be described through a user-defined wave spectrum or wave element, or through a time history of wave elevation. Numerous wave trains and directional wave spreading are allowed.
iii. Wind: Can be persistent, demonstrated by a wind spectrum or specified through a time history file. Generally, the wind is the same at all horizontal locations through the model [19].

Current and wave water particle velocities are described from the user-mentioned current profile and wave attributes, permitting for a single, nominal depth value in the wave estimation. OrcaFlex is a universal modelling tool and so these water particle velocities are normally unaltered by the existance of local variations or model objects in the seabed. However, it is possible to describe an area of disturbed water that is linked to an OrcaFlex vessel using a property called sea state RAOs (these data are depicted within the vessel type) [18]. The degree of disturbance must be represented by the user either by hand or by direct import from an OrcaWave or WAMIT diffraction output. A foregoing assessment of the wind and wave resources as well as their integration, at potential wind farm installation locations is critical as it can help distinguish the effect of the environmental conditions on the performance of the wind turbines, the supplement of their fatigue mechanical loads as an outcome of the hydrodynamic interaction with the oceanic waves at that particular location or even improve maintenance procedures. On that account, favorable locations for the installation of wind farms based on the development of historical meteorological data could be selected. Environmental features, for instance water depth, at the location of installation also influence the platform designs. There are multiple locations where fixed wind is unprofitable and/or technically impossible but where spar platforms would be too deep when upholding a large (10 MW+) turbine. With the drift towards larger turbines, it is likely that spars will not be acceptable in these locations and hence shallower platform designs are originating (TLP, semi-sub and barge). Mooring also highly relies on location as it is strongly dominated by water depth and the seabed conditions. Another instance of specialization to a specific location is an increased influence from port requirements, mostly as wind turbine sizes increase.[4] Standard ports are shallow, which stops deep spars from being built unless they are towed horizontally. In a floating wind turbine system, there is higher potential for motion of the support structure, which united with an absence of aerodynamic damping in the side-to-side direction, may cause wave, current, and wind directionality to more massively influence on fatigue loading [29]. Sea density, sea temperature, viscosity are kept constant. The conditions which are variable can be seen in table 6.

Table 6: Importance of analyzing different environmental conditions.

Environmental Conditions analyzed	Importance to analyze them
Wave Height	Affects the magnitude of stresses on the structure
Wave Period	Influences the frequency of stress cycles
Wave Speed	Affects the dynamic response of the structure
Wave Direction	Impacts the orientation of stresses on the structure
Current Speed	Influences the drag forces and vortex-induced forces
Current Direction	Impacts the orientation of current induced forces
Wind Speed	Affects the aerodynamic forces on the structure
Wind Direction	Impacts the orientation of wind induced forces

4.2 MISALIGNMENT

In floating offshore wind turbines (FOWT), misalignment happens due to the integrated effects of wind movement and wave direction as shown in Fig 14 . The control system of the wind turbine regulates the yaw angle based on wind direction, while waves cause alterations in the inclination of the turbine mast, changing the relative incidence angle of wind onto the rotor-blade plane [14].

An examination was held based on the impact of wind/wave misalignment on a floating wind turbine situated on a barge platform and discovered that the roll, yaw, and sway Response Amplitude Operators (RAOs) accelerated with misaligned wind and waves, mainly for a misalignment of 90º. Ramachandran et al.2 investigated the significance of wind/wave misalignment on a tension-leg platform (TLP) platform under wind excitation of 10 m/s and conveyed the significance of 90º and 180º misalignments in both the and the time series and spectral energy contents of the platform motions. A 1/34.5-scale paradigm of a wind turbine with a spar-buoy platform in an ocean engineering basin was modelled and were able to reconstruct storm situation with blades fully feathered using aligned wind, waves, and current and misaligned wind and waves. This research shows the influence of wind/wave misalignment on floating platforms, mainly in the side-to-side direction caused by higher compliance of the floating support structure [19].

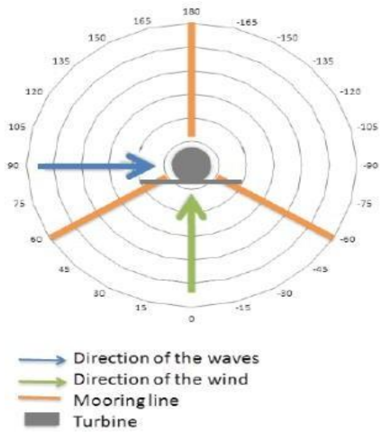


Figure 14: *A schematic showing the wind is fixed at 0 degree, so the wave propagation direction is equal to the wind/wave misalignment.[14]*

5. CASE STUDIES

5.1 5MW SPAR FOWT: MOORING LINE ANALYSIS

Turbine Characteristics

This example considers modelling of the NREL 5MW baseline turbine organized on the OC3 Hywind spar, is complex in nature and contains functionality to model the gearbox, generator, hub, blades and connected control systems. Each blade profile is given at a separate arc length along the blade length and the defined arc lengths are totally independent of how the blade is distributed into sections. The blade pitch, torque root connection moment and aero rotor power are analyzed. The turbine's response to various loads such as winds and waves is simulated using time domain analysis. Load analysis is performed for turbines ultimate loads estimation that ensures structural integrity [31]. The characteristics of the turbine can be seen in appendix 1.1.

Platform Characteristics

The Spar platform is an axisymmetric surface-penetrating buoy and is designed using the spar buoy category of 6D buoy in OrcaFlex. The layout of the spar platform consists of three important sections. The spar has a draft of 120m in its static position; meaning that the top of the spar, upon which the tower is organized, enlarges to an elevation of 10m above the still water level (SWL). The spar buoy structure is divided into a number of discrete cylinders. To precisely model surface-penetrating effects, a fine discretisation of 1m is allocated to the top section, tapered section, and the top 8m of the bottom section. For the rest of the bottom section, a discretisation of 10m is advised. The surge, sway, heave dynamics and roll, yaw and pitch rotations are analyzed [31]. The characteristics of the platform can be seen in appendix 1.2.

Mooring Characteristics

The floating system is anchored to the seabed by three different mooring lines shown in Fig 15. Each mooring line is allocated suitable physical characteristics via the corresponding line type. The spar anchors and mooring lines are located proportionately around the platform, in azimuth supplement of 120°, with the anchor radius approximately 854m from the middle of the spar platform. To establish an element of yaw stiffness to the platform, individual mooring line is linked to the spar by means of a 'crowfoot' (delta) connection. Every leg of the delta connection (component A & B) is allocated a length and target segment length of 30m & 1m, respectively. Furthermore, end A of each line is linked to the buoy at a fixed radius of 5.2m from the spar's central axis. The moorings are composed of steel and fiber reinforced polymers. The effective tension, acceleration, velocity of moorings are analyzed [31]. The characteristics of the mooring can be seen in appendix 1.3.

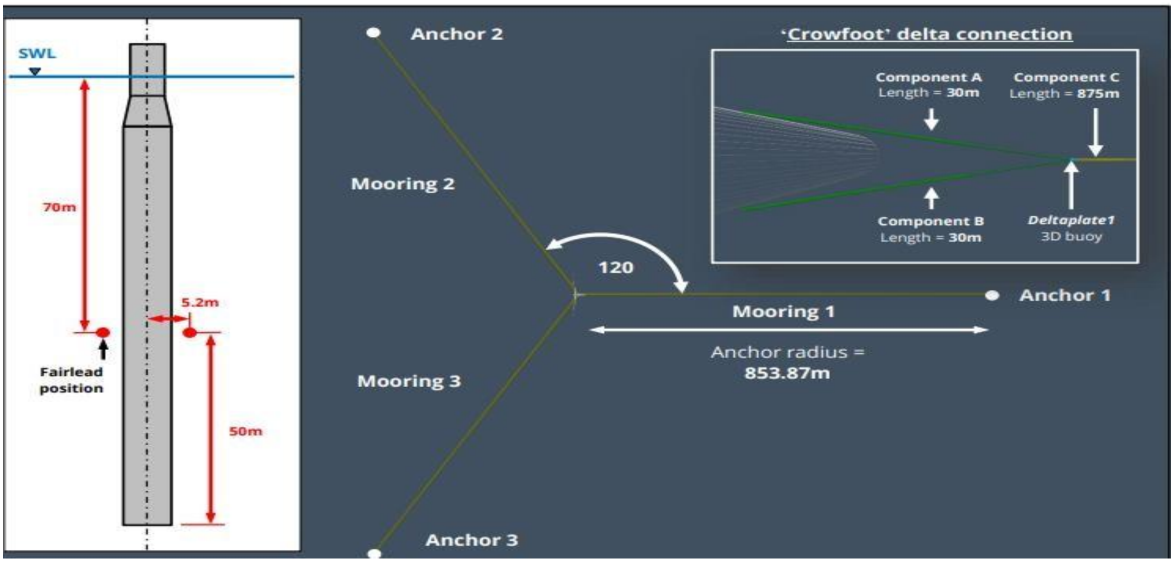


Figure 15: *A schematic diagram showing different components of mooring lines and also the spar platform .[31]*

5.2 15MW SEMI-SUBMERSIBLE FOWT: MOORING LINE ANALYSIS

<u>Turbine Characteristics</u>

The turbine rotor represents version 1.1.3 of the 15MW reference wind turbine (RWT). The turbine takes the configuration of a three-bladed rotor with variable-speed and collaborative blade-pitch control capabilities. The turbine hub is linked to the nacelle, which houses a direct-drive synchronous generator. The nacelle is held by the tower which is organised upon the University of Maine (UMaine) VolturnUS-S reference semi-submersible platform. The turbine's response to various loads such as winds and waves is simulated using time domain analysis. Load analysis is performed for turbines ultimate loads estimation that ensures structural integrity. The blade pitch, torque root connection moment and aero rotor power are analyzed [32]. The characteristics of the turbine can be seen in appendix 1.4.

<u>Platform Characteristics</u>

The semi-submersible steel platform model consists of three 12.5m-diameter outer columns linked to a 10m-diameter central column along three rectangular pontoons. As shown in Fig 16. The material of the platform is steel or hybrid type (steel and composites). For the column pontoon connection, it is bolted and has an X-bracing system. The surge, sway, heave dynamics and roll, yaw and pitch rotations are analyzed. The surge, sway, heave dynamics and roll, yaw and pitch rotations are analyzed [32]. The characteristics of the platform can be seen in appendix 1.5.

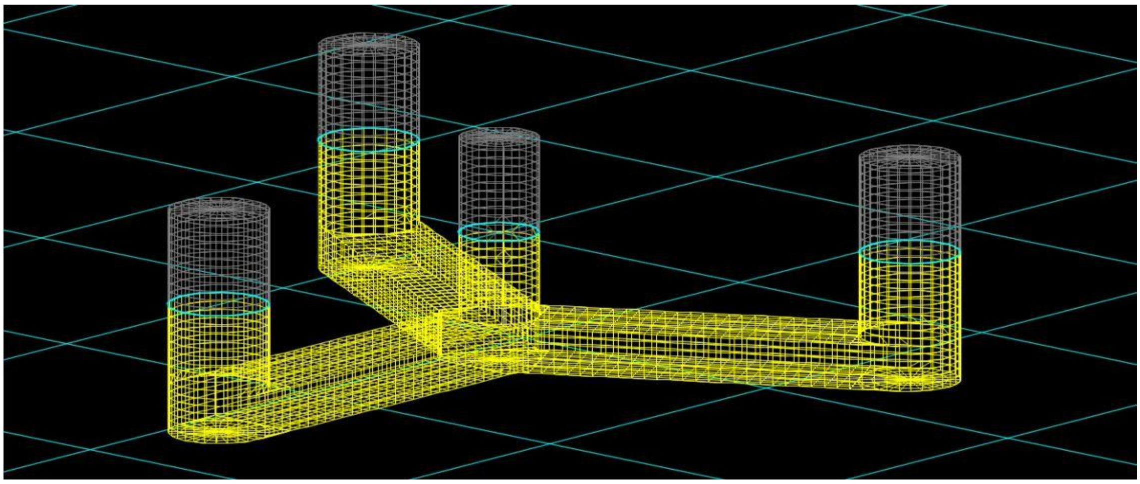


Figure 16: *A schematic diagram is showing the semi-submersible platform structure .[32]*

Mooring Characteristics

The moorings have a 3–4-line catenary configuration. They have asymmetrical layouts. The connectors and fittings are mostly constructed using aluminum or titanium (corrosion resistant). The mooring lines are made of steel. The effective tension, acceleration, velocity of three different moorings is analyzed [32]. The characteristics of the mooring can be seen in appendix 1.6.

5.3 10MW BOTTOM FIXED OWT: BLADE ANALYSIS

Turbine Characteristics

This example models the 10MW reference wind turbine (RWT), developed as part of the International Energy Agency's (IEA) Wind Task 37.The turbine hub is linked to the nacelle, which houses a direct-drive synchronous generator. The nacelle is held by a tower which is organized upon a fixed-bottom monopile base implanted to a depth of 45m below the seabed as shown in Fig 17. The turbine support structure consists of a Tower and Monopile, designed using line objects. The corresponding line types use the homogeneous pipe category, which permits for designing of the variable outer and inner diameter profiles of these items as well as their physical characteristics. The turbine's response to various loads such as winds and waves is simulated using time domain analysis. Load analysis is performed for turbines ultimate loads estimation that ensures structural integrity. The blade pitch, torque root connection moment aero rotor power and generator power are analyzed [33]. The characteristics of the turbine can be seen in appendix 1.7.

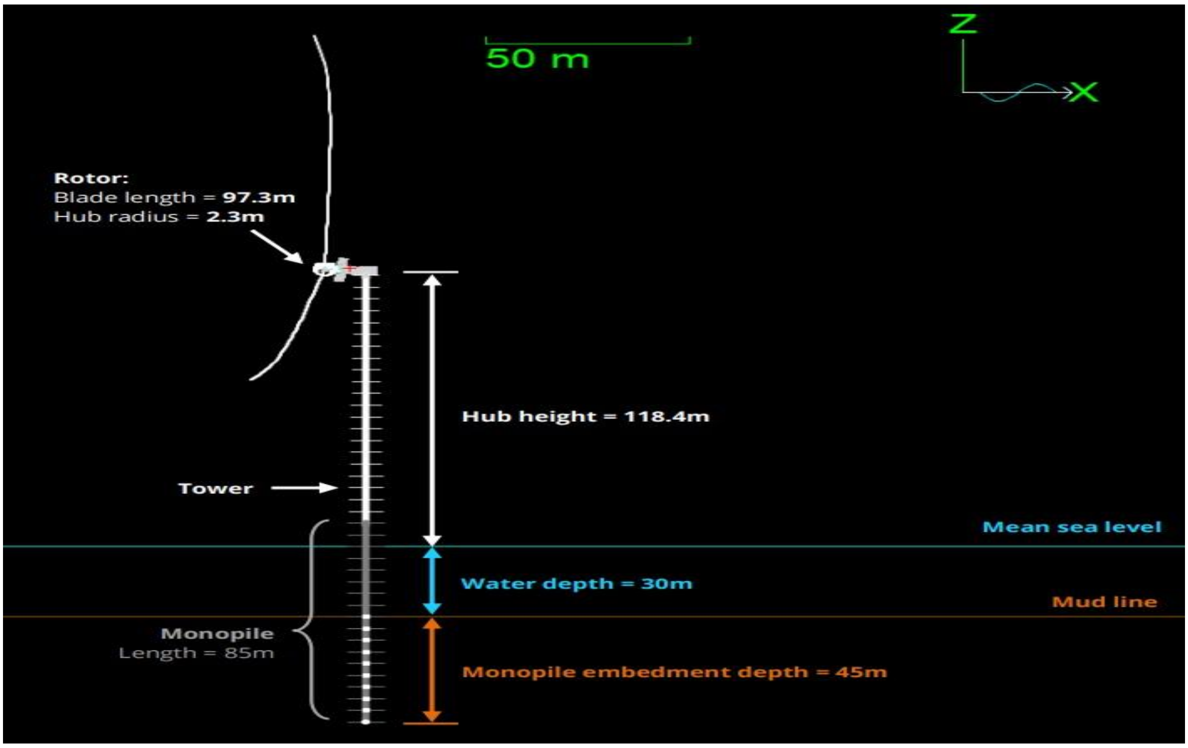


Figure 17: *A schematic diagram is showing different sections of turbine and tower.[33]*

6. RESULTS

Different parameters of mooring lines and turbine of 5MW, 10MW, 15MW scales and variable platforms are plotted against a simulation time of 1800 seconds. Fig 20-32 shows the relationship of different attributes with time. The environmental conditions considered here are wind speed 8m/sec, wave height 3m and wave period 12 seconds. A wind speed of 8m/sec is a moderate one suitable for testing turbine performance in realistic scenarios. A wave period of 12 seconds is common in typical locations for offshore wind farms and a wave height of 3m is a moderate wave height and a well representative of offshore conditions. The sea elevation is measured in m and wind speed in m/sec. Fig 18-19 shows the wind speed and wave characteristics over time.

6.1 5MW FOWT SYSTEM DYNAMICS

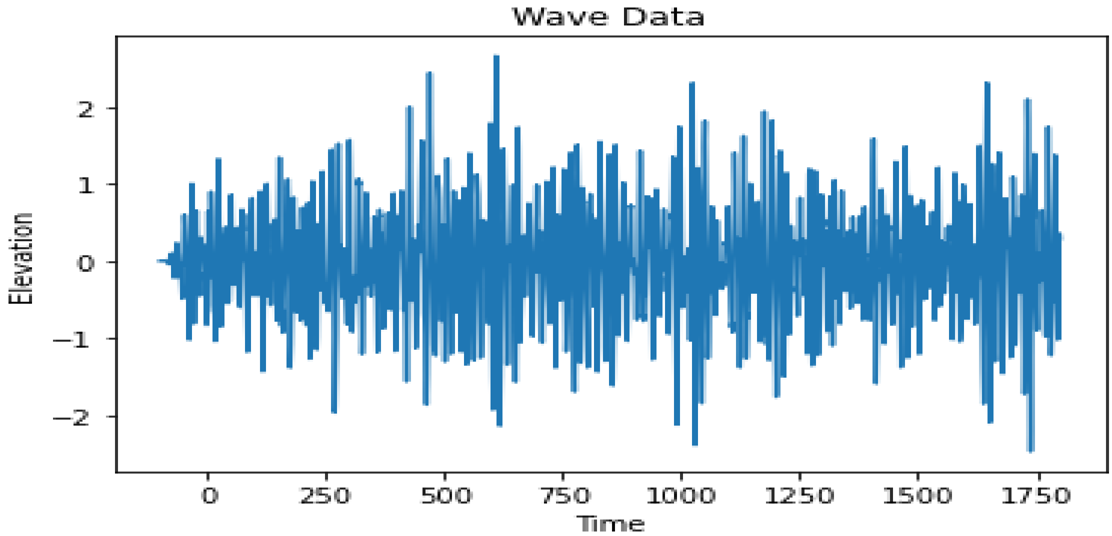


Figure 18: *Sea elevation changing with time*

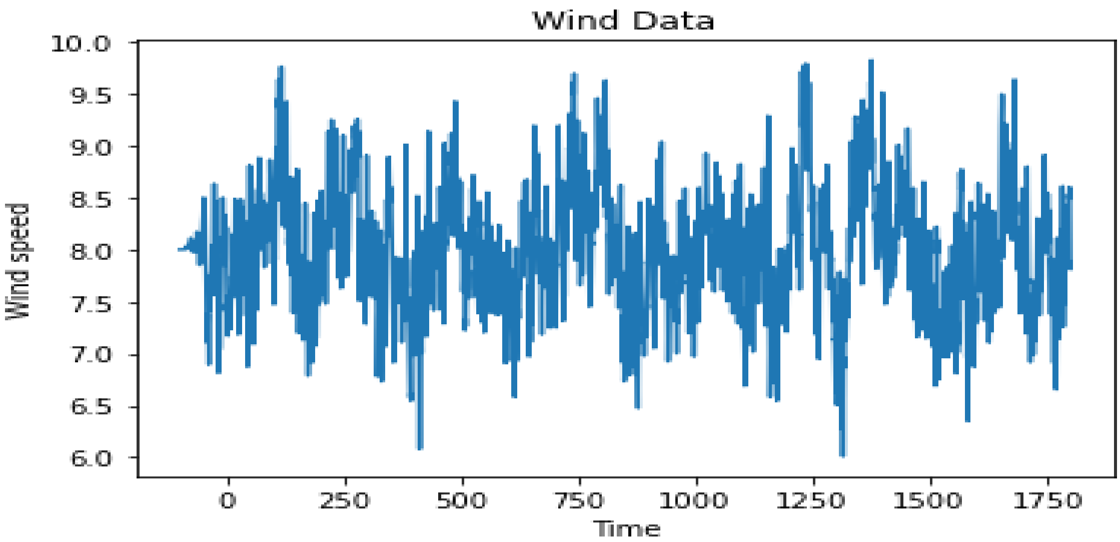


Figure 19: *Wind speed changing with time*

In Fig 20 the effective tension is measured in kN. The effective tension of mooring system shows increasing variability over time. Higher wind speeds cause increased tension variability and greater range in tension. The system appears more dynamic and responsive in later periods because of prolonged exposure to environmental forces.

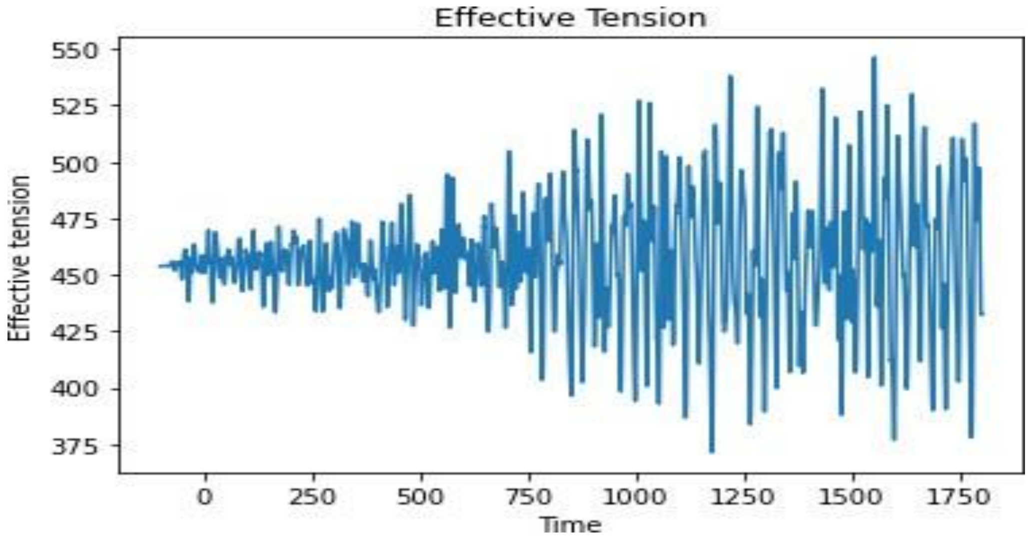


Figure 20: *Effective tension of mooring lines*.

In Fig 21 the acceleration is measured in m/sec^2. Due to increase in wave height the acceleration increases. Strong wind speeds (upto 9-9.5m/sec) often precede periods of higher acceleration. Acceleration peaks are frequently aligned with sudden changes in tension. The acceleration shows immediate responses to environmental forces. This suggests acceleration could be a leading indicator of stress in mooring.

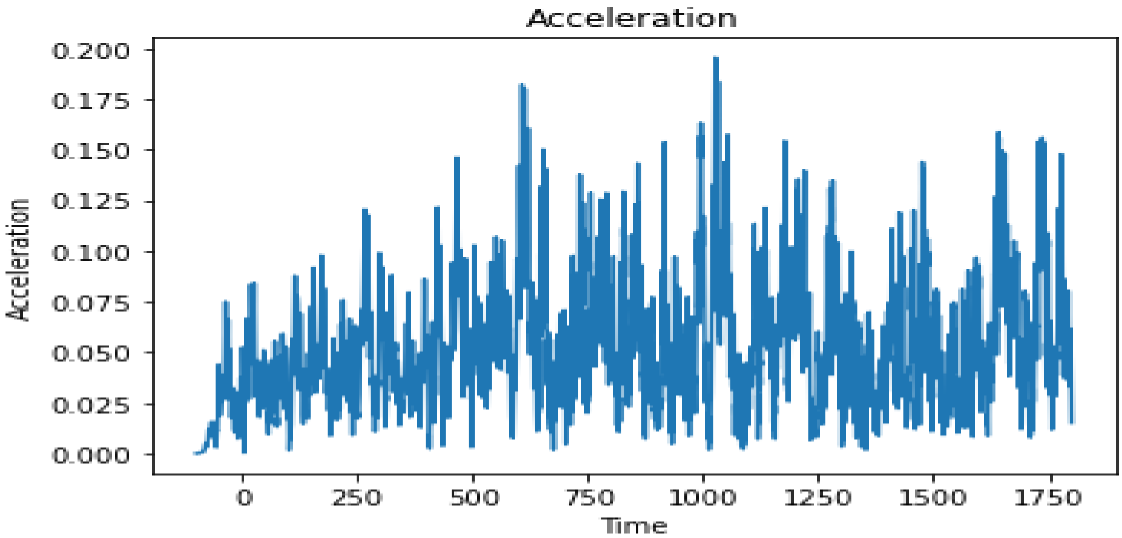


Figure 21: *Acceleration of mooring lines*.

In Fig 22 the dynamics (surge) is measured in m. There is a drift in the platform after 500 secs which cause a decrease in surge motion. This happened due to high thrust forces hitting the platform. This trust force also increases tension of the moorings. There is a visible damping of oscillations. An interaction between platform and surrounding water can exhibit nonlinear behaviour such as nonlinear wave loading which can sometime decreases the surge.

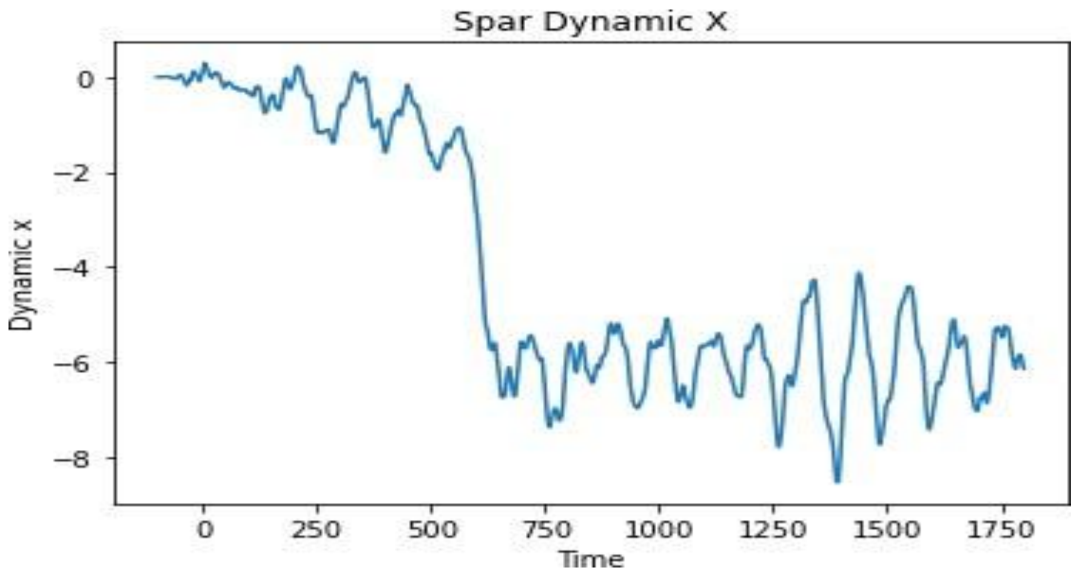


Figure 22: *Dynamics of Spar platform*.

In Fig 23 the torque is measured in kNm. At the beginning the torque starts increasing as rotor accelerates from rest. There is an exponential increase in the torque during the period where there is a major drop in surge. With increase in wind speed torque increases as the blades are producing more lift. The torque suggests wind turbine is responding to increasing wind speed, reaching operational range and maintaining power production despite variable conditions.

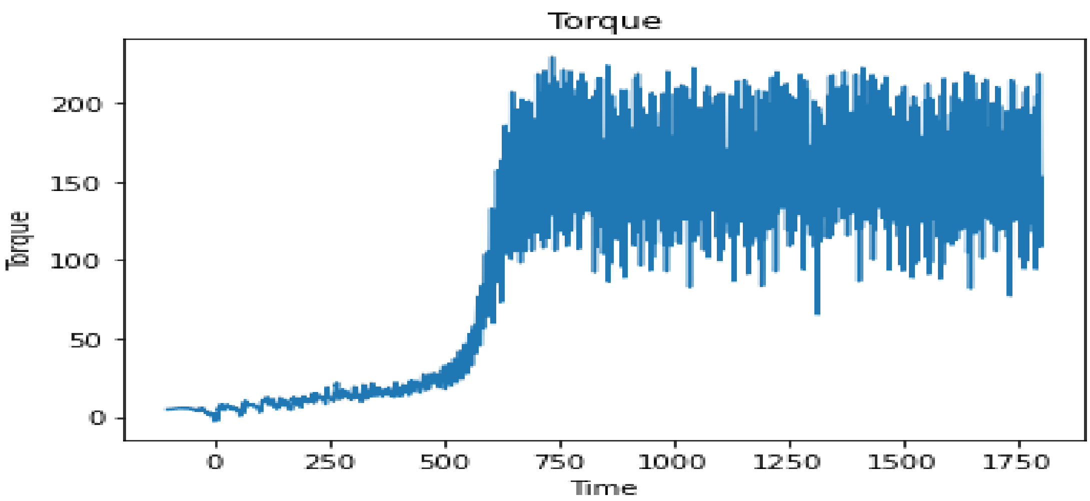


Figure 23: *Rotor torque of 5MW wind turbine*.

6.2 15MW FOWT SYSTEM DYNAMICS

In Fig 24 the acceleration is measured in m/sec^2. There is consistent oscillation pattern in later portion. There is an initial spike which represents significant environmental forces while the stabilization of oscillations shows normal operating conditions of mooring lines. The consistent pattern shows regular movements or stress on the moorings. It helps in monitoring structural health and dynamic behaviour of mooring system.

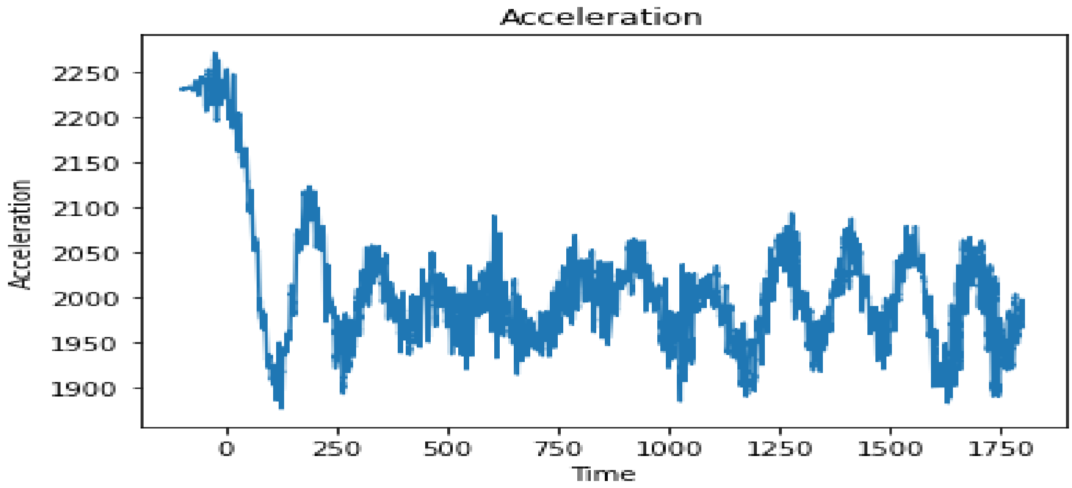


Figure 24: Acceleration *of mooring lines*.

In Fig 25 the effective tension is measured in kN. Along with increase in wind speed and wave height the effective tension also increases. As compared to 5MW the effective tension for 15MW mooring lines is much higher. The average tension is higher compared to initial state suggests that the system has settled to operational configuration.

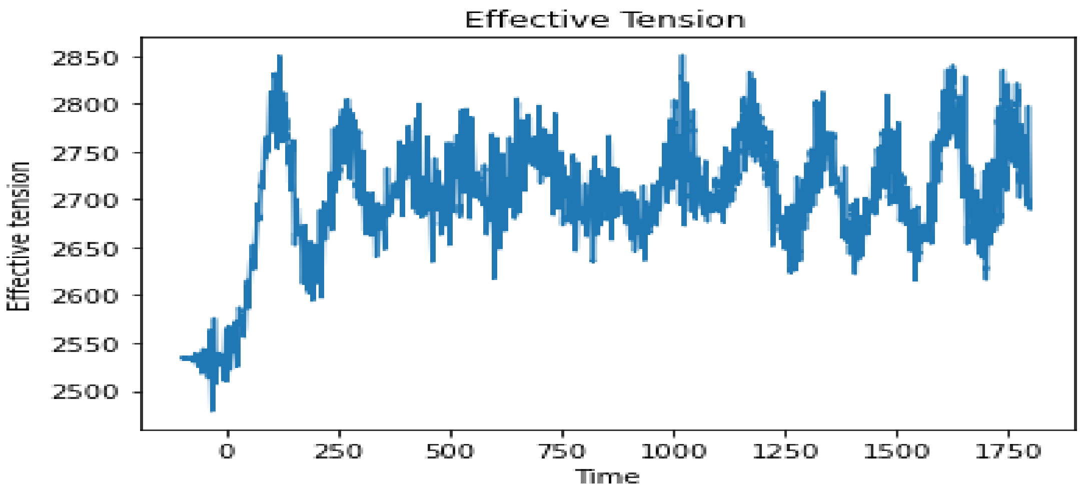


Figure 25: *Effective tension of mooring lines*.

In Fig 26 the dynamics (surge) is measured in m. There is a sharp decline in the first stage and then the motion settles into regular oscillating pattern. The motion is consistent in later stage causing regular wave like pattern. There is cyclic pattern that represents the platform's response to wave and wind forces while consistent nature of oscillations suggests the platform is maintaining stable operational parameters.

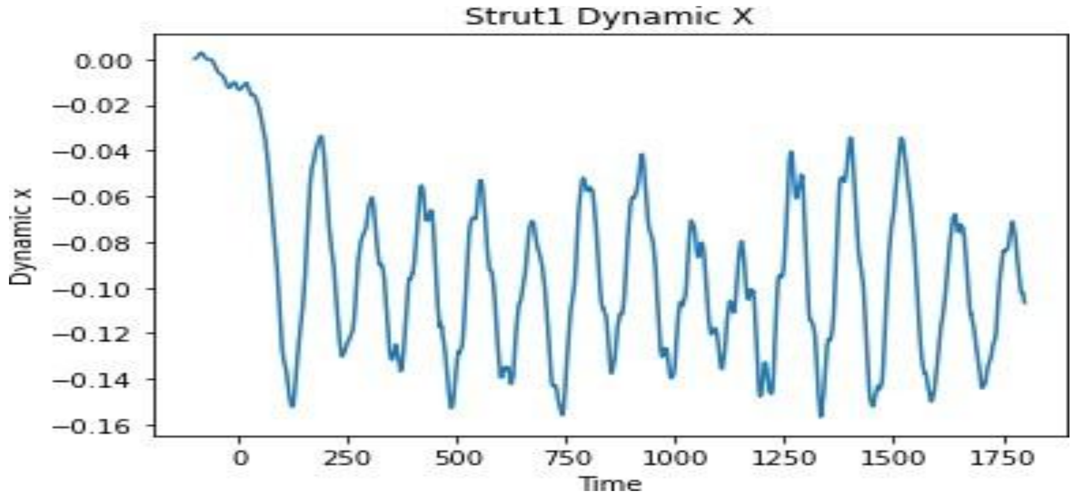


Figure 26: *Dynamics of Semi-Submersible platform*

In Fig 27 the torque is measured in kNm. The torque pattern suggests there is a long-term stability after the system quickly recovers from an initial negative spike. The platform maintains stable rotational forces which is crucial for operational safety of platform. The small ongoing oscillations likely to represent the platform's continuous dynamic response to environmental forces like wind and wave.

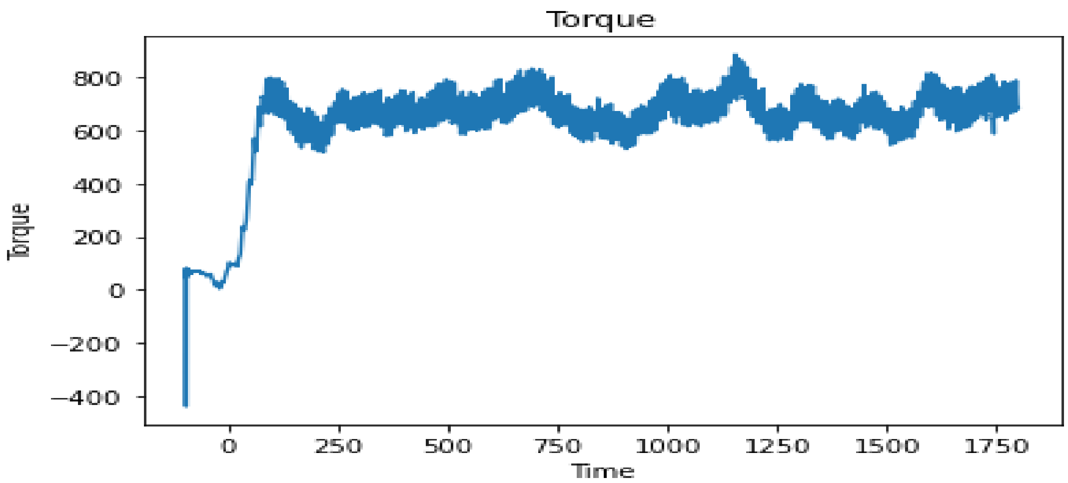


Figure 27: *Rotor torque of 15MW wind turbine*.

6.3 10MW OWT SYSTEM DYNAMICS

In Fig 28 the torque is measured in kNm. The torque vs time graph shows a sinusoidal curve. This is because as the rotor rotates, the blades experiences varying ange of attacks that causes change in lift and drag forces causing sinusoidal curve. Here no significant damping is taking place. This type of regular oscillating torque is important for power output stability.

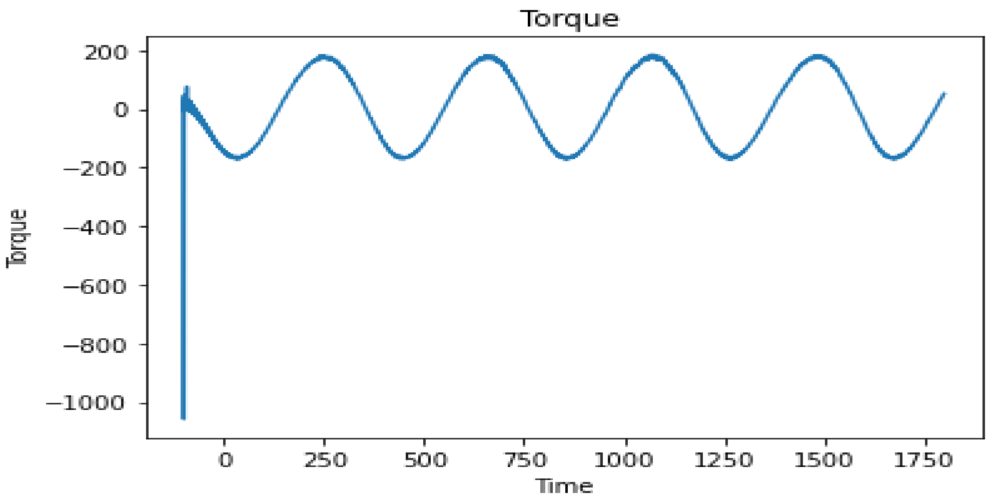


Figure 28: *Rotor torque of the 10MW wind turbine*.

In Fig 29 the root connection moment is measured in kNm. There are fluctuations in root connection moment over time and shows a periodic pattern with peaks and valleys. The root connection moment is influenced by wind and wave conditions that can create cyclic loading on turbine foundation. Other turbine, platform and mooring result graphs can be seen in appendix 2.1-2.6

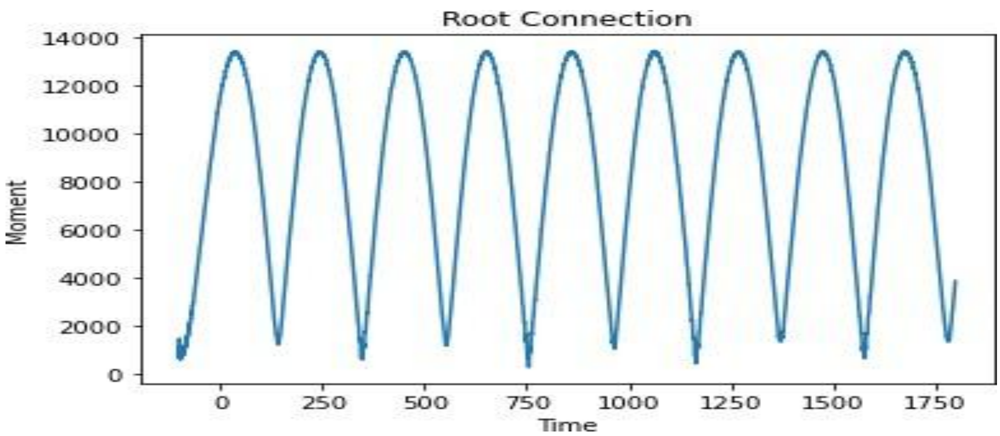


Figure 29: *Root Connection Moment in the blades of 10MW wind turbine.*

In Fig 30-39 the graphs of the statistical data from the mooring lines, turbine and buoy attributes are plotted against the environmental conditions and the logarithmic trendline is obtained along with the R^2 value to check which attributes are more prone to the environmental conditions. The statistical analysis has reduced 27 excel sheets of 27 cases into 1 post processing excel sheet.

6.4 5MW FOWT STATISTICAL ANALYSIS RESULTS

For a 5MW spar offshore wind turbine the wave period and wave height have more impact on the velocity of the moorings than wind speed. This is since the spar type often employs catenary or taut leg mooring systems which are sensitive to wave induced motion. This can also reduce tension variation. In Fig 30 the R squared value is mentioned along with the trendline that shows velocity is highly influenced by wave height.

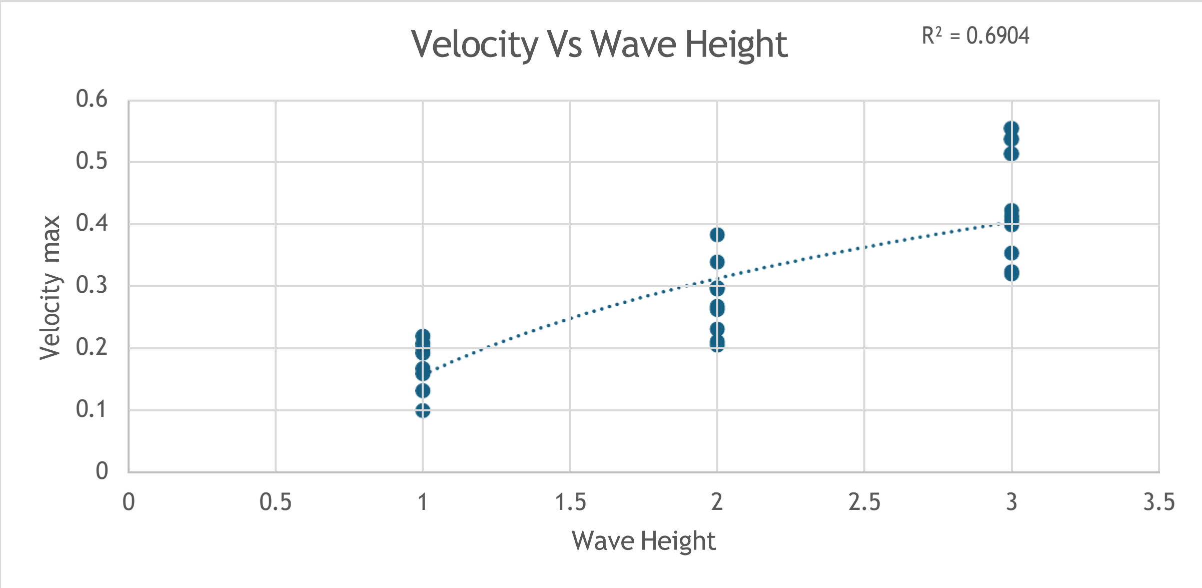


Figure 30: *Significant Influence of wave height on velocity of moorings.*

For a 5MW spar offshore wind turbine longer the wave period, it transfers more energy to the platform that causes higher accelerations. The spar platform's natural frequency is 10-15 secs (during simulation this range was considered). When wave periods approach this value, the platform resonates causing higher acceleration. In Fig 31 the R squared value is mentioned along with the trendline that shows acceleration is highly influenced by wave period.

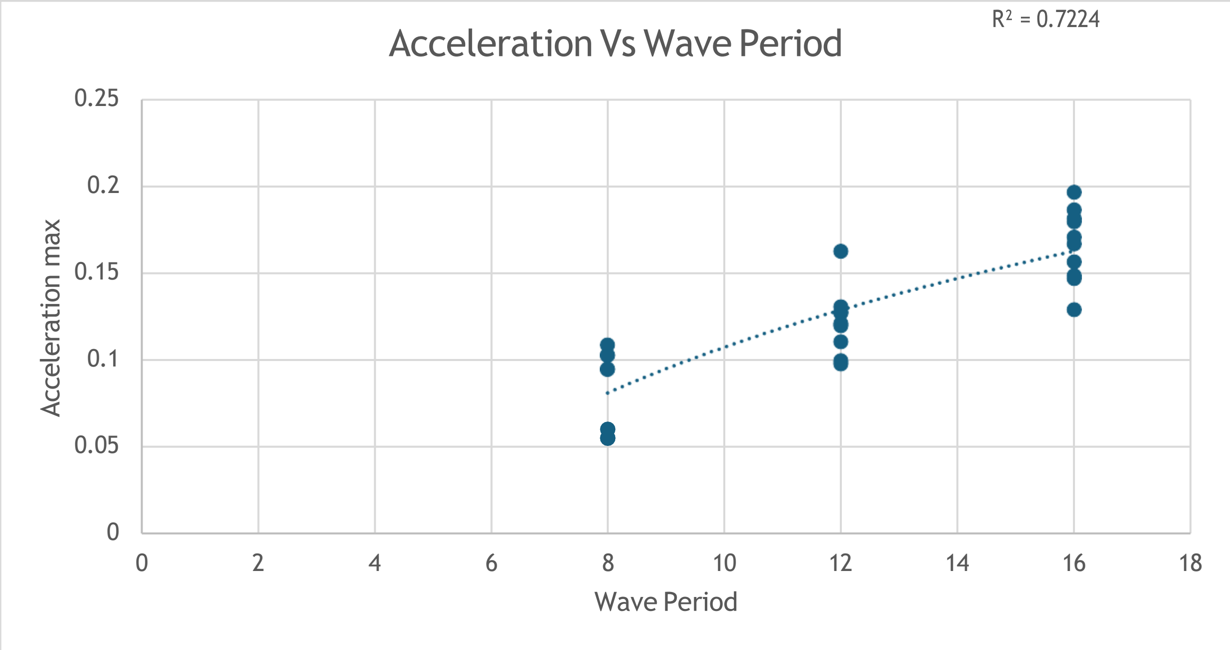


Figure 31: *Significant Influence of wave period on acceleration of moorings.*

For a 5MW spar offshore wind turbine the effective tension is less affected as the tension is a result of integrating forces over time making it less sensitive to short term fluctuations. The wave heights are relatively low as a result the platform responds slowly to smaller waves causing lower effective tension. In Fig 32 the R squared value is mentioned along with the trendline that shows effective tension is not much influenced by wave height.

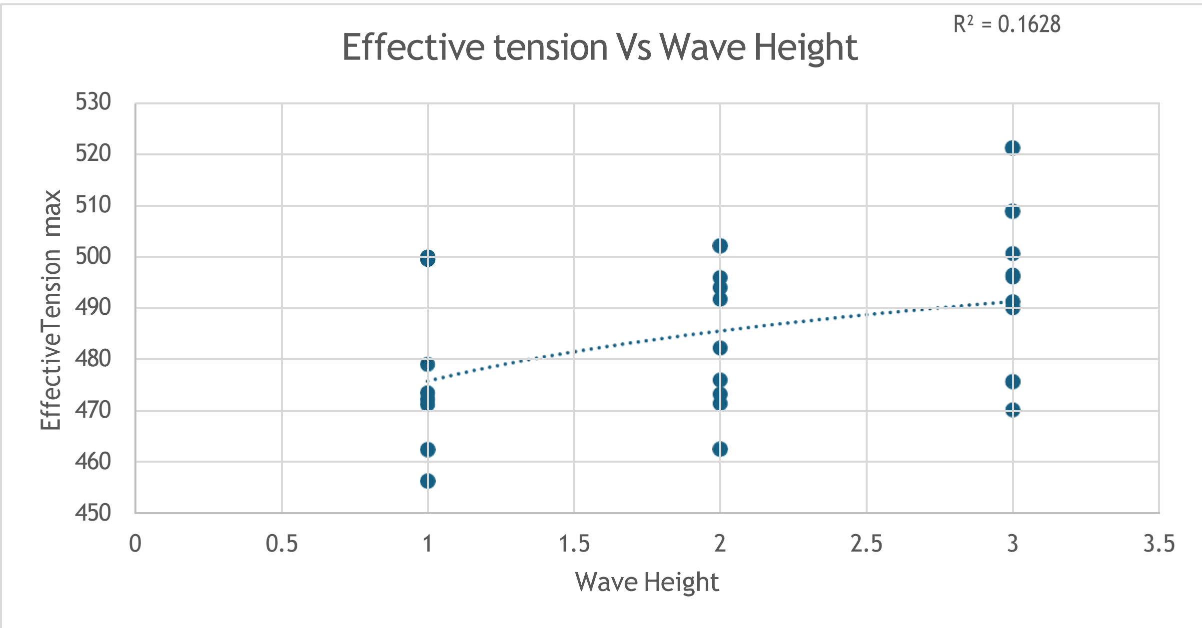


Figure 32: *Influence of wave height on effective tension of moorings.*

For a 5MW spar offshore wind turbine the surge dynamics is more affected by the wave height and wave period than wind speed. This is because the spar type has a large draft and small waterplane area making it sensitive to wave induced motions. But with a relatively lower wave height the movement in the platform gets slower and the surge motion decreases. In Fig 33 the R squared value is mentioned along with the trendline that shows surge motion is slightly influenced by smaller wave height, but the impact is more than on heave or sway motion.

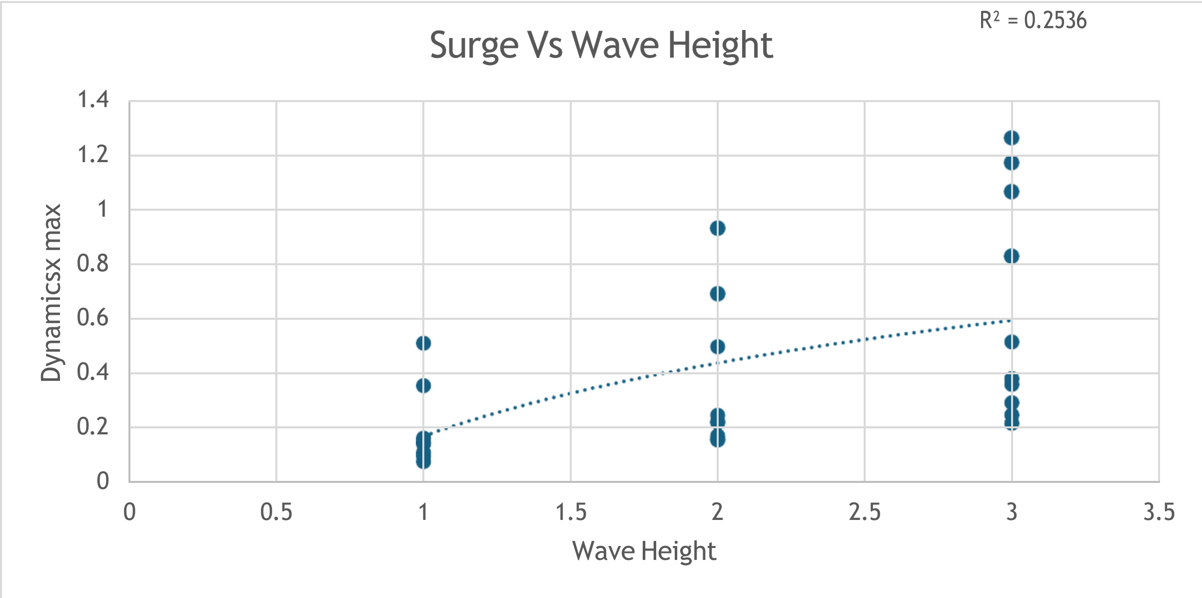


Figure 33: *Influence of wave height on surge motion of buoy.*

For a 5MW spar offshore wind turbine the root connection moment is affected by wind speed, wave height and wave period. Wave height has the highest influence as the wave induced foundation motion affects root connection moment. In Fig 34 the R squared value is mentioned along with the trendline that shows root connection moment is slightly influenced by lower wave heights.

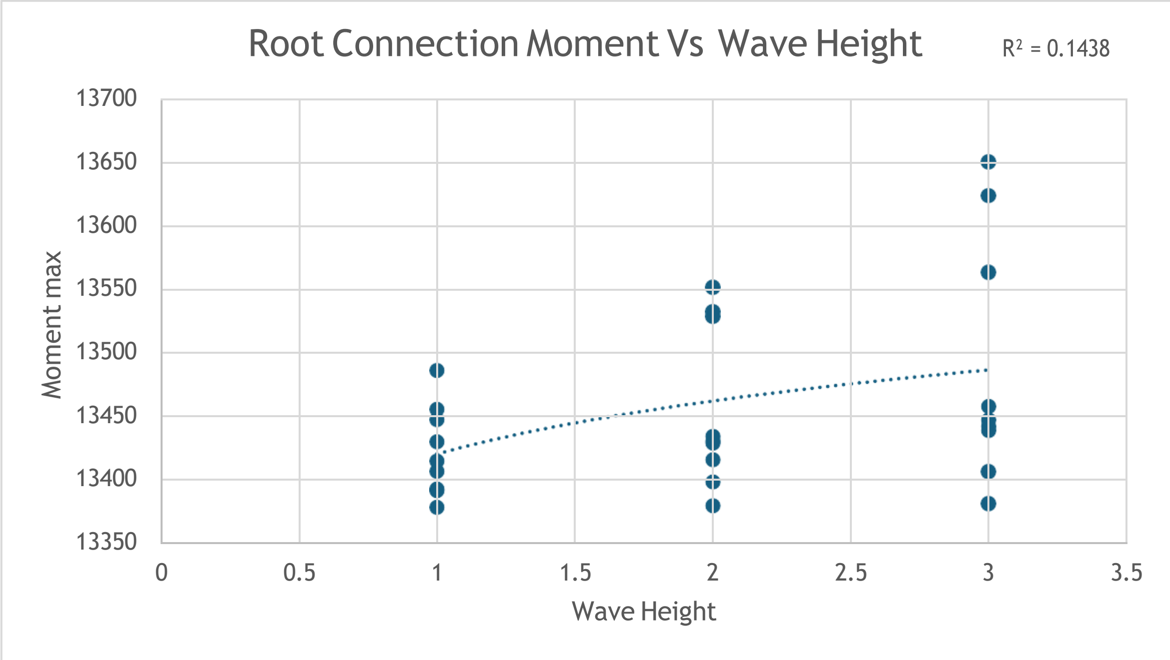


Figure 34: *Influence of wave height on root connection moment of turbine.*

6.5 15MW FOWT STATISTICAL ANALYSIS RESULTS

For 15MW semi-submersible offshore wind turbine velocity and acceleration is more sensitive to environmental conditions than the effective tension. The velocity is highly affected due to wave wind and current fluctuations. As the wind loading increases the velocity also increases. The platform and mooring design also influence the velocity. In Fig 35 the R squared value is mentioned along with the trendline that shows the velocity of mooring is highly influenced by wind speed.

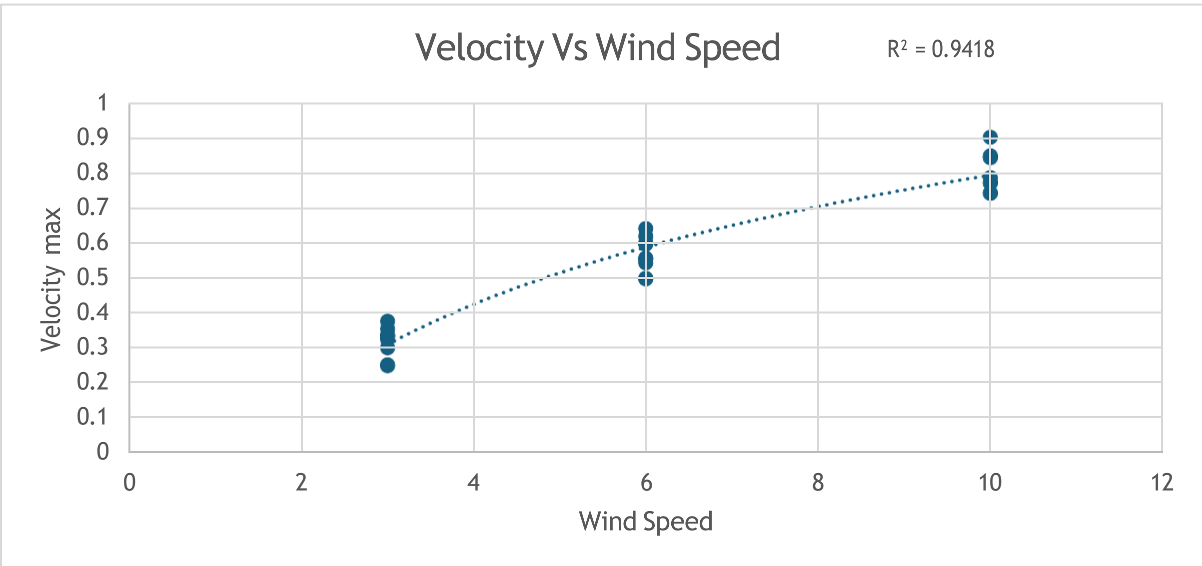


Figure 35: *Significant Influence of wind speed on velocity of moorings.*

For 15MW semi-submersible offshore wind turbine velocity and acceleration is more sensitive to environmental conditions than the effective tension. The dynamic environmental loading affects acceleration. In Fig 36 the R squared value is mentioned along with the trendline that shows the acceleration of mooring is highly influenced by wave height.

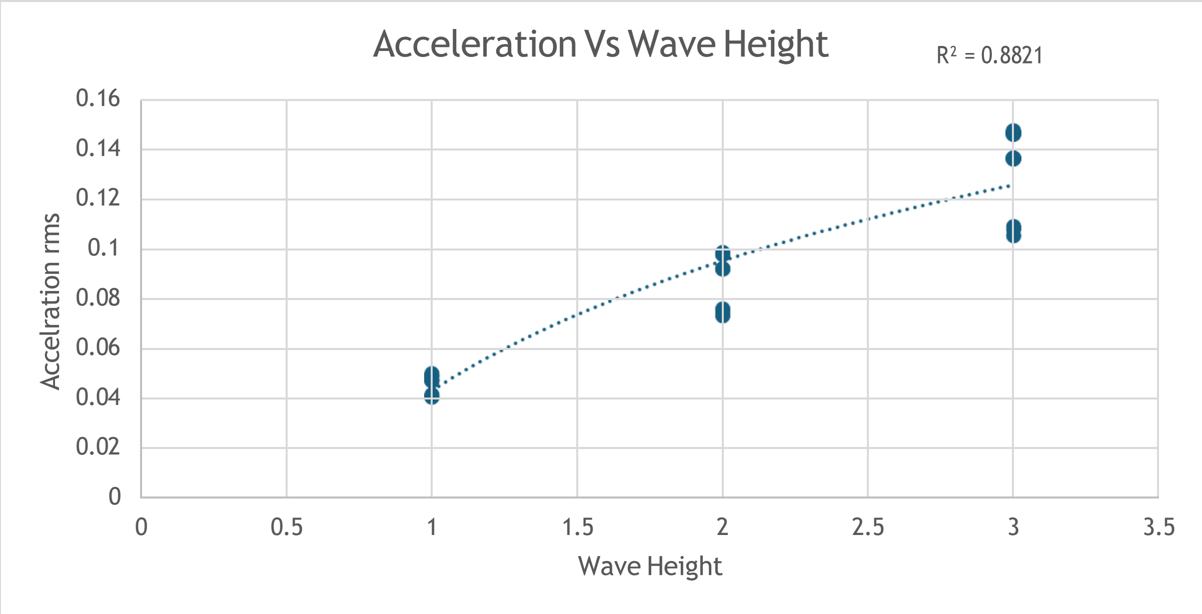


Figure 36: *Significant Influence of wind height on acceleration of moorings.*

The sway dynamics of a 15 MW wind turbine are more susceptible to environmental conditions than surge and heave motion. This is due to water flow impacts stability exacerbating sway motions. The wave induced forces increases sway. In Fig 37 the R squared value is mentioned along with the trendline that shows the sway motion g is more influenced by wave period than the heave and surge motion.

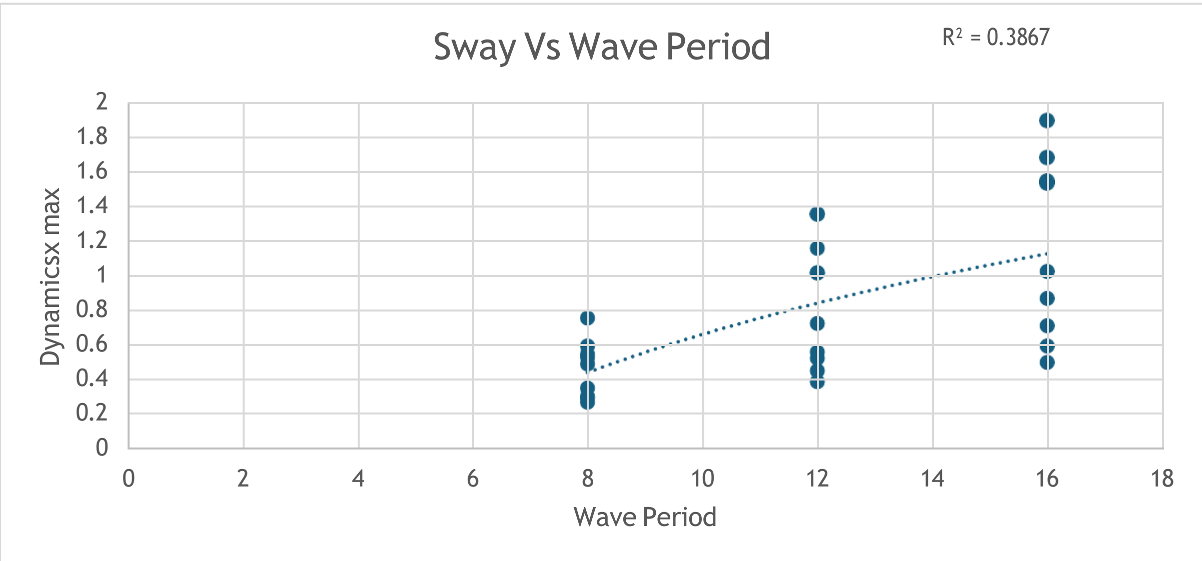


Figure 37: *Influence of wave period on sway motion of buoy.*

6.6 10MW OWT STATISTICAL ANALYSIS RESULTS

For a 10MW bottom fixed turbine root connection moment is highly influenced by wave height, the wind speed. Wave height has the highest influence as the wave induced foundation motion affects root connection moment. Here the influence of environment on root connection moment seems very less as design's structural stiffness resists external load caused by environment and reduces turbine's deflection and rotations which reduces the impact on root connection moment. In Fig 38 and in Fig 39 the R squared value is mentioned along with the trendline that shows root connection moment is very slightly influenced by lower wave heights.

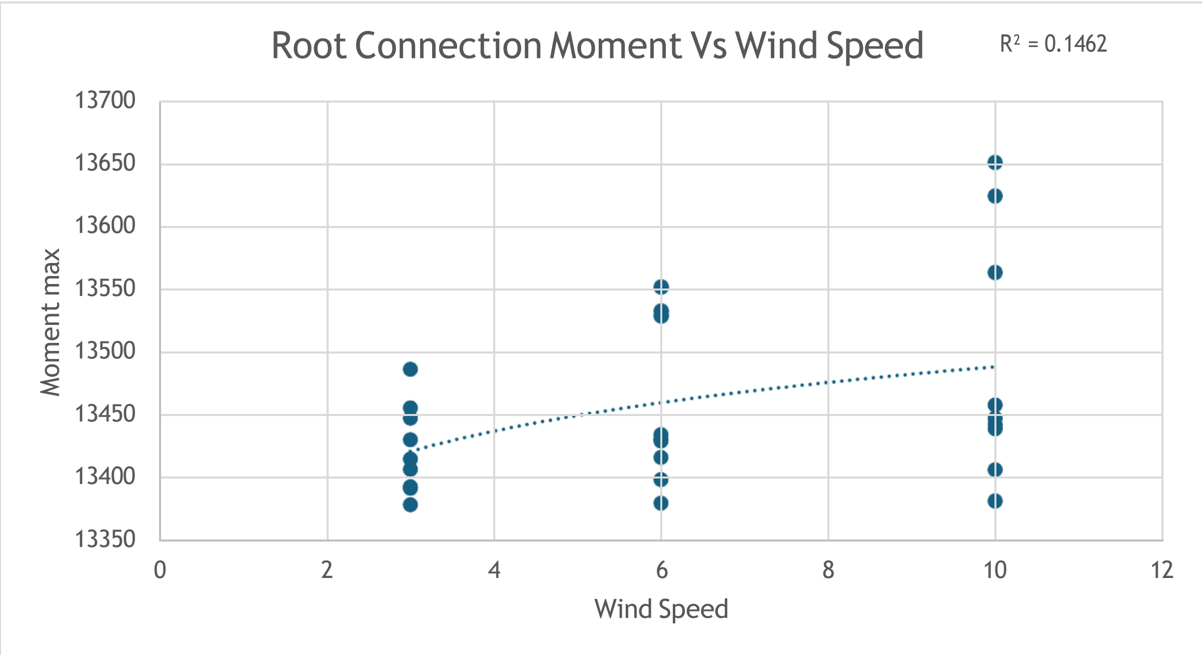


Figure 38: *Influence of wind speed on root connection moment of turbine.*

Bottom fixed foundations are mostly monopiles, jackets or gravity bases structures which gives a rigid connection to seabed. The rigidity minimizes the influence of environment on turbine's structural response. Other statistical results graphs can be seen in appendix 2.7-2.14.

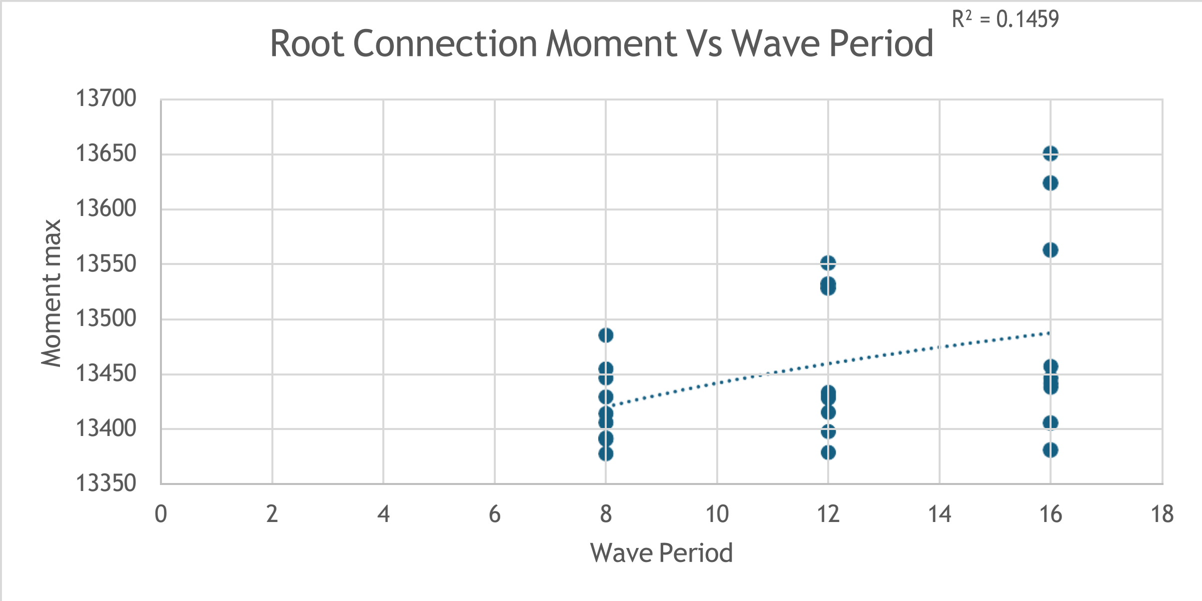


Figure 39: *Influence of wave period on root connection moment of turbine.*

6.7 FATIGUE ANALYSIS 5MW SPAR

The table below (Table 7) shows the rainflow damage (0.00223), the total damage of all load cases 0.045 and the lifetime of the mooring lines is 21.94 years. The gamma value is used in fatigue calculation when offshore wind turbines are subjected to cyclic loading. The k and grade parameters are used to characterize the fatigue behavior of materials. The realization and probability are used to quantify the uncertainty and variability of fatigue related parameters. Rainflow damage is used to calculate the cumulative damage caused by variable amplitude loading on a component or structure. The damage_RFC refers to rainflow cycle counting which is used to calculate the damage accumulation of a component or structure under variable amplitude loading. The damage is measured in MPa.

Table 7: *This table represents the rainflow damage, total damage of all the load cases and the life for 5MW Spar Floating Offshore Wind Turbine.*

Critical Characteristics	***Values obtained during Analysis***
Gamma	*1.3293*
k	*3.0*
Grade	*R3*
C	*1200000000.0*
Number of Cases	*27*
Probability	*1*
D_rainflow	*0.002333*
Total_Damage	*0.0455*
Lifetime	*21.94 years*
Wave_height	*[1,2,3]*
Wave_period	*[8,12,16]*
Wind_speed	*[3,6,8]*
Component	*Stud chain*

6.8 FATIGUE ANALYSIS 15MW SEMI-SUBMERSIBLE

The table below (Table 8) shows the rainflow damage (0.00333), the total damage of all load cases 0.064 and the lifetime of the mooring lines is 15.52 years. In 15MW imposes higher loads on moorings due to larger size. The mooring line of 15MW wind turbine has larger diameter and subjected to higher loads that leads to increased fatigue damage as a result the lifetime of 15MW is less than the 5MW wind turbine mooring system. The graphs representing fatigue damage of moorings for both 5MW and 15MW can be seen in appendix 2.15-2.16.

Table 8: *This table represents the rainflow damage, total damage of all the load cases and the life for 15MW Semi-Submersible Floating Offshore Wind Turbine.*

Critical Characteristics	***Values obtained during Analysis***
Gamma	*1.3293*
k	*3.0*
Grade	*R3*
C	*120000000000.0*
Number of Cases	*27*
Probability	*1*
D_rainflow	*0.00333*
Total_Damage	*0.064*
Lifetime	*15.52 years*
Wave_height	*[1,2,3]*
Wave_period	*[8,12,16]*
Wind_speed	*[3,6,8]*
Component	*Stud chain*

7. DISCUSSION & AREA OF IMPROVEMENT

Floating offshore wind turbines are exposed to harsh environmental conditions affecting their reliability, lifespan and performance. The findings provide valuable insights into the design and optimization of offshore wind turbines. The 5MW Spar FOWT has moderate acceleration and velocity whereas the effective tension is slightly less as the platform is more stable which reduces platform motion and subsequent effective tension. The dynamics and rotation of platform is lower compared to the other scales of wind turbines. For 15MW semi-submersible FOWT the acceleration and velocity of mooring is significantly high dure to floating structure. There Is increase in dynamics and platform rotations than the 5MW spar FOWT. The root connection moment of 15MW is more than 10MW and 5MW as the turbine has a stiffer tower to support higher loads which increases the moment (7000kNm for 5MW, 14000kNm for 10MW and 22000kNm for 15MW), it has an increased rotor diameter which results in a higher moment arm and increased loads on tower base. It is more sensitive to wind and wave alignment compared to 10MW and 5MW wind turbines. The 10MW is less sensitive to wave and wind alignment because of its bottom fixed structure. The fatigue damage of 15MW semi-submersible FOWT is more than the 5MW Spar type because of increased motion hence have a lower lifespan. The 5MW has a higher life span. The fatigue analysis highlights the significance of considering mooring system configuration when evaluating fatigue damage and lifetime of mooring lines. The results provide valuable insights for optimizing mooring systems to minimize fatigue damage and extend lifetime. As the scale of wind turbine increases from 5MW to 15MW the effective tension of the moorings increases (550kN to 2850kN) as larger turbines produce more thrust force that leads to increased tension in moorings as well as velocity and acceleration also increases. The torque in the rotor, root connection moment, and rotor aero power also increases with the scale. As the rotor diameter increases with scale it captures more wind energy leading to higher torque. The length of the blade increases which leads to higher lift in blade and increase in torque. The root connection moment increases with the scale The area of improvement includes technical improvement (advanced numerical models, coupled dynamics), methodological improvement (comparison with existing studies, uncertainty analysis), environmental conditions improvement (extreme events, season variations), design optimization improvements (cost-benefit analysis, material selection) and software tools improvement (data visualization, custom scripting).

8. CONCLUSIONS AND RECOMMENDATION

Among 5MW, 15MW and 10MW the 15MW semi-submersible is more affected by the environmental conditions. Semi-submersibles are mostly used in deeper waters where the environmental conditions are more severe. There is stronger currents larger waves and increased turbulence in deeper waters which leads to higher loads on the structure which later increases the fatigue damage and in turn decreases the lifetime. Semi-submersible has a complicated structure than the spar and bottom fixed designs with multiple columns pontoons and mooring lines. This complexity makes it more sensitive towards environment. Hydrodynamic loading from waves and currents on the platform can cause increased stress on structure. The wind and wave alignment current induced loads and wave current interactions all of them has a major impact on making the 15MW wind turbine more prone to environmental conditions. Spar designs have simpler structure and are not much affected by wave induced motions whereas the bottom fixed designs are mostly used in shallower waters where the severity of environment is less. The spar type is also less subjected to fatigue damage as a result the lifetime of the moorings are longer than the semi-submersible one. This also concludes that more the scale of the wind turbine more loading will it face which will increase the fatigue damage and finally result in decreased lifetime of the moorings. Some recommendations should be kept in mind like a combination of numerical and experimental methods should be conducted, multiple environmental conditions should be considered, suitable materials for moorings should be selected, redundancy in mooring system should be incorporated to ensure continued operation in case of failure in components. Rainflow cycle counting method should be used to accurately predict the fatigue, fatigue crack growth should be considered, Palmgren-miner rule can be utilized for estimating cumulative fatigue damage from multiple loading cycles.

9. FUTURE WORK

New materials should be explored to improve mooring system performance and reduce fatigue. More sophisticated numerical models should be created to capture complex interactions between mooring system, wind turbine and environmental condition. Full scale testing of mooring systems should be conducted to validate numerical models to improve understanding of real-world behaviour.

REFERENCES

[1] Azin Lamei, Masoud Hayatdavoodi, "Motion and elastic response of wind-tracing floating offshore wind turbines". Research Article Open Access Published: 01 September 2022 Volume 9, pages 43–67, (2023).

[2] Yunpeng Zhu, Jing Zhong, "Effects of the yaw error and the fault conditions on the dynamic characteristics of the 15 MW offshore semi-submersible wind turbine". Received 2 July 2023; Received in revised form 7 March 2024; Accepted 7 March 2024.Volume 300.Pages117440.

[3] Nouman Saeed , Jingliang Gong,"A novel design of multifunctional offshore floating platform structure based on topology optimization". Received 10 September 2023; Received in revised form 22 December 2023; Accepted 26 February 2024.Volume 306.Pages 117782

[4] Louise Efthimiou (World Forum Offshore Wind e.V.)Hayden Marcollo (AMOG/Moorsure) Chairman Cables & Floating Substations Subcommittee, "Floating Offshore Wind Manager Floating Offshore Wind Dynamic Cables: Overview of Design and Risks." March 2023.

[5] Francisco Pimenta, "Predictive model for fatigue evaluation of floating wind turbines validated with experimental data". Received 27 June 2023; Received in revised form 3 January 2024; Accepted 8 January 2024.Volume 223. Page 119981

[6] Vladimir Yakimov, Fang Wang, Fuxi Zhang & Rajiv Balakrishna, "Floating wind turbines structural details fatigue life assessment". Volume 13. Pages 16312 (2023).

[7] Izwan Bin Ahmad , Anja Schnepf , "An optimisation methodology for suspended inter-array power cable configurations between two floating offshore wind turbines". Received 5 October 2022; Received in revised form 13 March 2023; Accepted 30 March 2023.Volume 278.Pages114406.

[8] A. Campanile V. Piscopo, "Mooring design and selection for floating offshore wind turbines on intermediate and deep-water depths". Received 15 February 2017; Received in revised form 7 July 2017; Accepted 16 November 2017.Volume 148. Pages 349-360.

[9] Zhengru Ren , Hongyu Zhou , Binbin Li,"Localization and topological observability analysis of a moored floating structure using mooring line tension measurements". Received 6 May 2022; Received in revised form 19 September 2022; Accepted 23 September 2022.Volume 266, Part 5. Pages 112706.

[10] Salem Okpokparoro, "Reliability analysis of floating wind turbine dynamic cables under realistic environmental loads". Received 6 February 2023; Received in revised form 29 March 2023; Accepted 13 April 2023.Volume 278. Pages114594.

[11] Shengjie Rui, Zefeng Zhou, "A review on mooring lines and anchors of floating marine structures". Received 11 December 2023; Received in revised form 9 March 2024; Accepted 5 May 2024.Volume 199. Pages114547.

[12] Priya Saj, Sajith A S,"Dynamic and Fatigue Analysis of Mooring Lines of Iea 15mw Floating Offshore Wind Turbine in Normal and Accidental Condition". Preprint submitted to Journal of Ocean Engineering June 15, 2023. Pages 11,

[13] E. Land & Z. Hu, "The effects of surge underprediction on fatigue damage of floating offshore wind turbine dynamic cables". February 2023, DOI:10.1201/9781003399759-56. In book:

"Advances in the Analysis and Design of Marine Structures" (pp.503-510).

[14] Aitor Saenz-Aguirre, Alain Ulazia, "Floating wind turbine energy and fatigue loads estimation according to climate period scaled wind and waves". Received 23 July 2022; Received in revised form 27 September 2022; Accepted 28 September 2022.Volume 271. 116303.

[15] CHUANG Zhen-ju, LIU She-wen, "Dynamic Analysis of Semi-Submersible Production Platform Under the Failure of Mooring Lines". Received February 5, 2020; revised July 27, 2020; accepted August 20, 2020.Volume 35, pages 84–95.

[16] Xinyi Li, Liwei Yu, Shuqing Wang, "Numerical Simulation of Hybrid Synthetic Mooring Lines Based on a Finite Difference Dynamic Model". Paper No: OMAE2022-81238, V05BT06A025. Published Online: October 13, 2022.Volume 30510 pages.

[17] Xiangcou Zheng Mohammed Seaid, "A fully coupled dynamic water-mooring line system: Numerical implementation and applications". Received 14 April 2023; Received in revised form 9 January 2024; Accepted 16 January 2024.Volume 294. Article 116792.

[18] Zhitao Guo,Xudong Zhao, "Simulation Study on Methods for Reducing Dynamic Cable Curvature in Floating Wind Power Platforms". Submission received: 20 December 2023 / Revised: 5 February 2024 / Accepted: 7 February 2024 / Published: 15 February 2024. J. Mar. Sci. Eng. 2024, 12(2), 334.

[19] Xiaocui Chen,Qirui Wang, Qirui Wang, "Dynamic Behavior of a 10 MW Floating Wind Turbine Concrete Platform under Harsh Conditions". Submission received: 20 December 2023 / Revised: 21 January 2024 / Accepted: 23 January 2024 / Published: 26 January 2024. Volume and Pages 12, 412.

[20] Saeid Fadaei *ORCID,Fred F. Afagh, "A Survey of Numerical Simulation Tools for Offshore Wind Turbine Systems". Submission received: 16 August 2023 / Revised: 2 November 2023 / Accepted: 25 December 2023 / Published: 10 January 2024. doi: 10.20944/preprints202308.1345.v2.

[21] T. Kim, F.J. Madsen, "Numerical analysis and comparison study of the 1:60 scaled DTU 10 MW TLP floating wind turbine". Received 20 September 2022, Revised 15 November 2022, Accepted 18 November 2022, Available online 22 November 2022, Version of Record 30 November 2022.Volume 202. Pages 210-221.

[22] Sung Youn Boo Yoon-Jin Ha, "Concept Design of a 15 MW TLP-Type Floating Wind Platform for Korean Offshore Installation". Submission received: 12 March 2024 / Revised: 7 May 2024 / Accepted: 8 May 2024 / Published: 10 May 2024. Volume and Page 12, 796.

[23] Wenjie Zhong, Xiaoming Zhang, "Hydrodynamic characteristics of a 15 MW semi-submersible floating offshore wind turbine in freak waves". Received 14 February 2023, Revised 8 May 2023, Accepted 10 June 2023, Available online 21 June 2023.Volume 283. Article 115094.

[24] Le Zhou aXin Shen,"Unsteady aerodynamics of the floating offshore wind turbine due to the trailing vortex induction and airfoil dynamic stall" Received 10 January 2024, Revised 17 April 2024, Accepted 27 May 2024, Available online 28 May 2024, Version of Record 14 June 2024.Volume 304. Article 131845.

[25] Xiaojiang Guo, †,Yu Zhang, "Integrated Dynamics Response Analysis for IEA 10-MW Spar Floating Offshore Wind Turbine". Submission received: 11 February 2022 / Revised: 10 April 2022 / Accepted: 11 April 2022 / Published: 14 April 2022. Volume and Page10(4), 542.

[26] Zhixin Zhao Wei Shi, "Dynamic analysis of a novel semi-submersible platform for a 10 MW wind turbine in intermediate water depth". Received 5 May 2021, Revised 11 August 2021, Accepted 12 August 2021, Available online 18 August 2021, Version of Record 18 August

2021.Volume 237. Article 109688.

[27] Carlos Eduardo Silva de Souza, Erin E. Bachynski-Polić, "Design, structural modeling, control, and performance of 20 MW spar floating wind turbines". Received 19 October 2021, Revised 7 January 2022, Accepted 21 January 2022, Available online 4 March 2022, Version of Record 4 March 2022.Volume 84. Article 103182.

[28] Simon Lefebvre, Maurizio Collu, "Preliminary design of a floating support structure for a 5 MW offshore wind turbine". Received 11 March 2011, Accepted 3 December 2011, Available online 20 December 2011. Volume 40. Pages 15-26.

[29] Adebayo Ojo a, Maurizio Collu, "Multidisciplinary design analysis and optimization of floating offshore wind turbine substructures: A review". Received 27 May 2022, Revised 22 September 2022, Accepted 24 September 2022, Available online 7 October 2022, Version of Record 7 October 2022.Volume 266, Part 1. Article 112727.

[30] Zine Ghemar, "Navigating Fatigue Analysis with Rain flow Counting: A Practical Overview" Submitted: 04 January 2024 Accepted: 11 January 2024 Published: 22 January 2024. Nov Joun of Appl Sci Res 1(1), 01-06.

[31] https://www.orcina.com/resources/examples/?key=k ,K01 5MW spar FOWT.pdf

[32] https://www.orcina.com/resources/examples/?key=k ,K03 15MW semi-sub FOWT.pdf

[33] https://www.orcina.com/resources/examples/?key=k K02 10MW fixed-bottom OWT.pdf

[34] Eguzkiñe Martinez-Puente , Ander Zarketa-Astigarraga, "Benchmarking of spectral methods for fatigue assessment of mooring systems and dynamic cables in offshore renewable energy technologies" Received 16 January 2024, Revised 20 May 2024, Accepted 27 May 2024, Available online 7 June 2024, Version of Record 7 June 2024. Volume 308, 15 September 2024, 118311.

APPENDICES

1. Turbine, Platform and Mooring Summary

1.1 Table 1: Turbine Properties (5MW)

Diameter (m)	125.88		
Area (m^2)	12.45e3		
Weight (kN)	1078.73		
Mass (te)	110.0		
Centre of mass (m)	2.9e-15	3.3e-15	-0.45375
Mass moments of inertia (te.m^2)	19.55e3	9.1e-12	2.3e-12
	9.1e-12	19.55e3	-1.2e-12
	2.3e-12	-1.2e-12	39.06e3
Blade 1			
Weight (kN)	173.971		
Mass (te)	17.7401		
Centre of mass relative to root (m)	0.02071	0.14226	20.4751
Blade 2			
Weight (kN)	173.971		
Mass (te)	17.7401		
Centre of mass relative to root (m)	0.02071	0.14226	20.4751
Blade 3			
Weight (kN)	173.971		
Mass (te)	17.7401		
Centre of mass relative to root (m)	0.02071	0.14226	20.4751

1.2 Table 2: Platform Properties (5MW)

Weight (kN)	73.22e3		
Buoyancy (kN)	84.04e3		
Submerged weight (kN)	-10.82e3		
Mass (te)	7466.33		
Displaced mass (te)	8570.0		
Submerged mass (te)	-1103.67		
Centre of mass (m)	0.0	0.0	30.0845
Volume (m^3)	8360.97		
Centre of volume (m)	0.0	0.0	60.596
Mass radius of gyration (m)	23.8	23.8	4.69
Total contact area (m^2)	1169.8		
Number of vertices	320		
Contact area per vertex (m^2)	3.65562		

1.3 Table 3: Mooring Properties (5MW)

Total length (m)	30.0
Total weight excluding contents (kN)	11.4306
Total weight including nominal contents (kN)	11.4306
Total buoyancy (kN)	0.48142
Total submerged weight including nominal contents (kN)	10.9492
Total mass excluding contents (te)	1.1656
Total mass including nominal contents (te)	1.1656
Total displaced mass (te)	0.04909
Total submerged mass including nominal contents (te)	1.11651

1.4 Table 4: Turbine Properties (15MW)

Diameter (m)	241.677		
Area (m^2)	45.87e3		
Weight (kN)	2692.96		
Mass (te)	274.605		
Centre of mass (m)	13e-15	230e-18	-1.6213
Mass moments of inertia (te.m^2)	175.1e3	21e-12	-3.3e-12
	21e-12	175.1e3	40e-12
	-3.3e-12	40e-12	346.8e3
Blade 1			
Weight (kN)	670.923		
Mass (te)	68.4151		
Centre of mass relative to root (m)	0.01718	0.40783	27.3723
Blade 2			
Weight (kN)	670.923		
Mass (te)	68.4151		
Centre of mass relative to root (m)	0.01718	0.40783	27.3723
Blade 3			
Weight (kN)	670.923		
Mass (te)	68.4151		
Centre of mass relative to root (m)	0.01718	0.40783	27.3723

1.5 Table 5: Platform Properties (15MW)

Weight (kN)	0.0		
Buoyancy (kN)	264.772		
Submerged weight (kN)	-264.772		
Mass (te)	0.0		
Displaced mass (te)	26.9992		
Submerged mass (te)	-26.9992		
Centre of mass (m)	0.0	0.0	20.25
Volume (m^3)	26.3407		
Centre of volume (m)	0.0	0.0	20.25
Total contact area (m^2)	36.855		
Number of vertices	40		
Contact area per vertex (m^2)	0.92137		

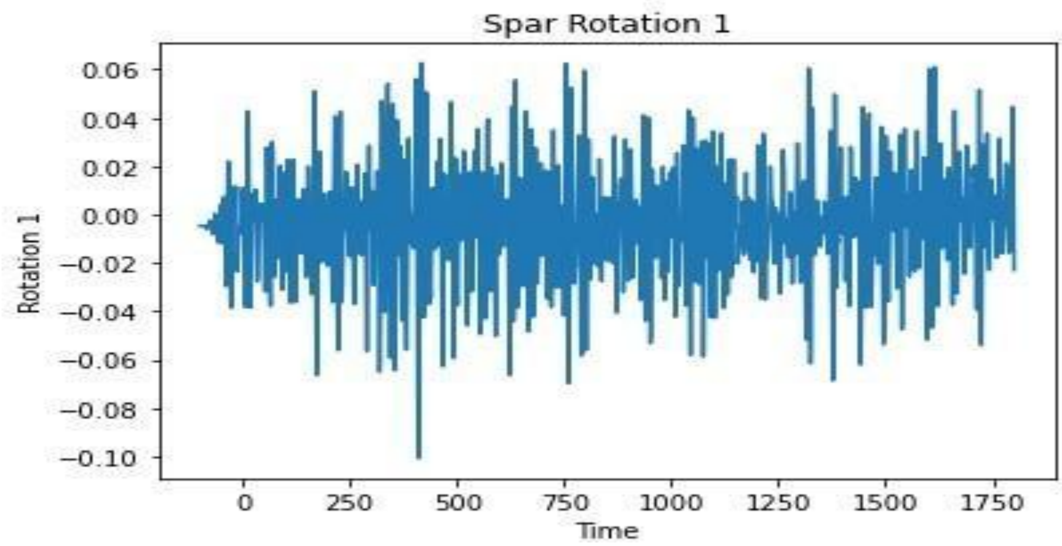
1.6 Table 6: Mooring Properties (15MW)

Total length (m)	850.0
Total weight excluding contents (kN)	5709.92
Total weight including nominal contents (kN)	5709.92
Total buoyancy (kN)	745.563
Total submerged weight including nominal contents (kN)	4964.36
Total mass excluding contents (te)	582.25
Total mass including nominal contents (te)	582.25
Total displaced mass (te)	76.0263
Total submerged mass including nominal contents (te)	506.224

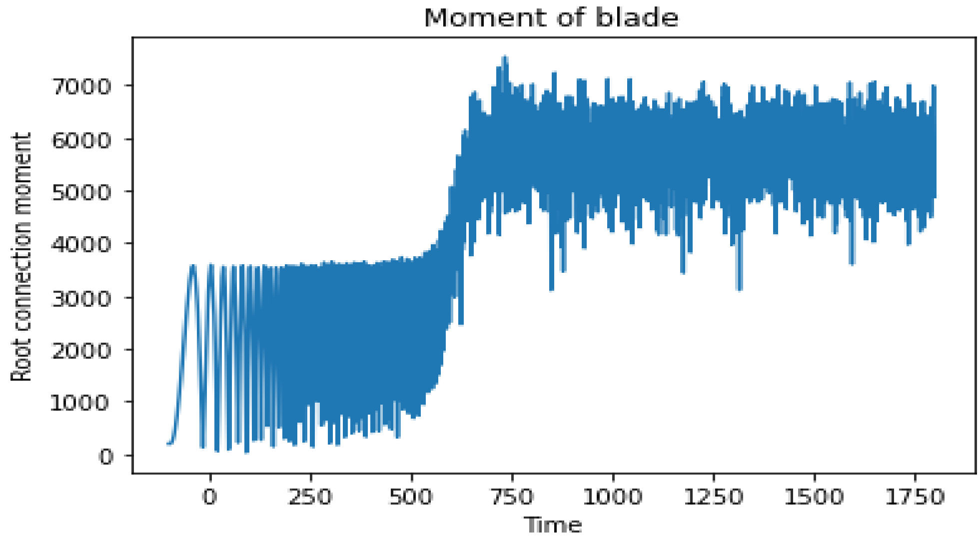
1.7 Table 7: Turbine Properties (10MW)

Diameter (m)	198.697		
Area (m^2)	31.01e3		
Weight (kN)	2232.26		
Mass (te)	227.627		
Centre of mass (m)	2.5e-15	4.5e-15	-1.60094
Mass moments of inertia (te.m^2)	108.9e3	-77e-12	750e-15
	-77e-12	108.9e3	12e-12
	750e-15	12e-12	214.9e3
Blade 1			
Weight (kN)	476.996		
Mass (te)	48.6401		
Centre of mass relative to root (m)	-0.35872	0.14756	28.0744
Blade 2			
Weight (kN)	476.996		
Mass (te)	48.6401		
Centre of mass relative to root (m)	-0.35872	0.14756	28.0744
Blade 3			
Weight (kN)	476.996		
Mass (te)	48.6401		
Centre of mass relative to root (m)	-0.35872	0.14756	28.0744

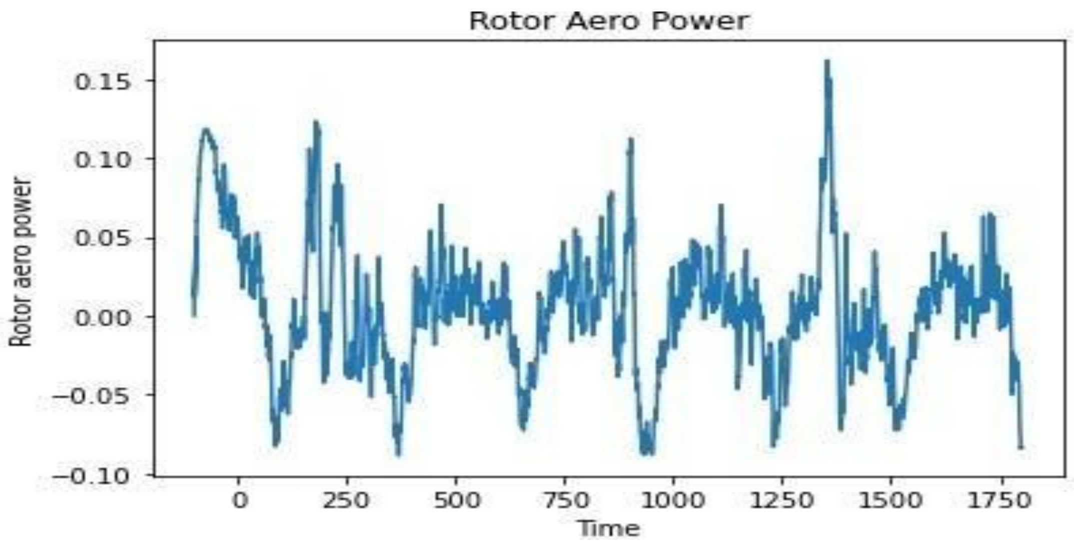
1. Other Results



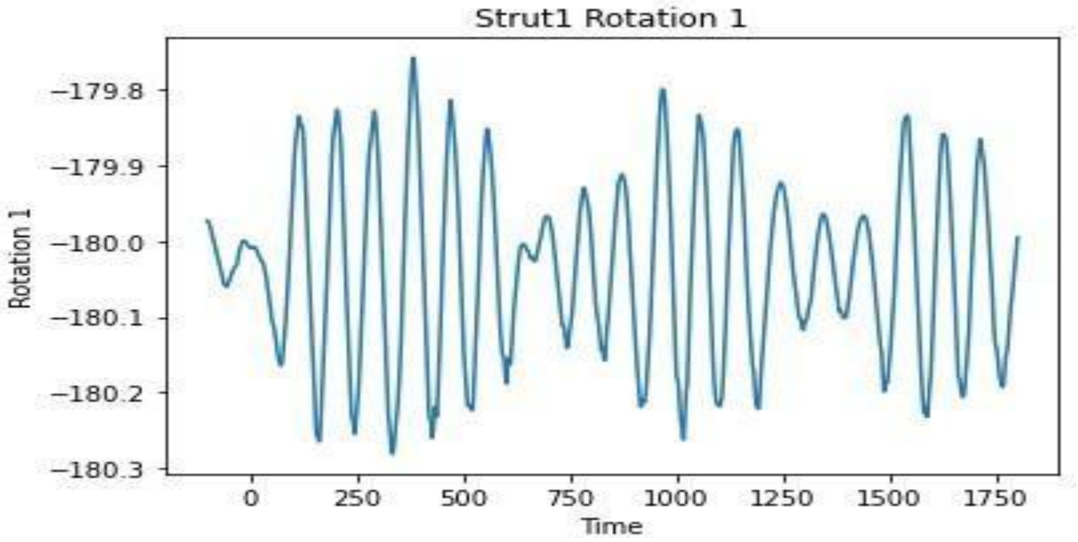
2.1 Figure 1: Rotation of Spar platform (5MW).



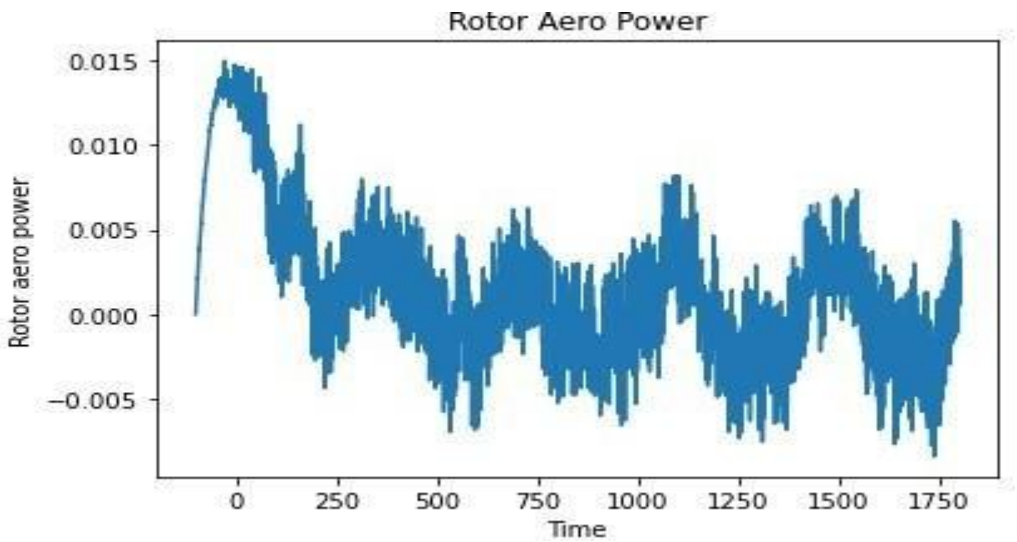
2.2 Figure 2: Root connection moment in blades of the 5MW wind turbine.



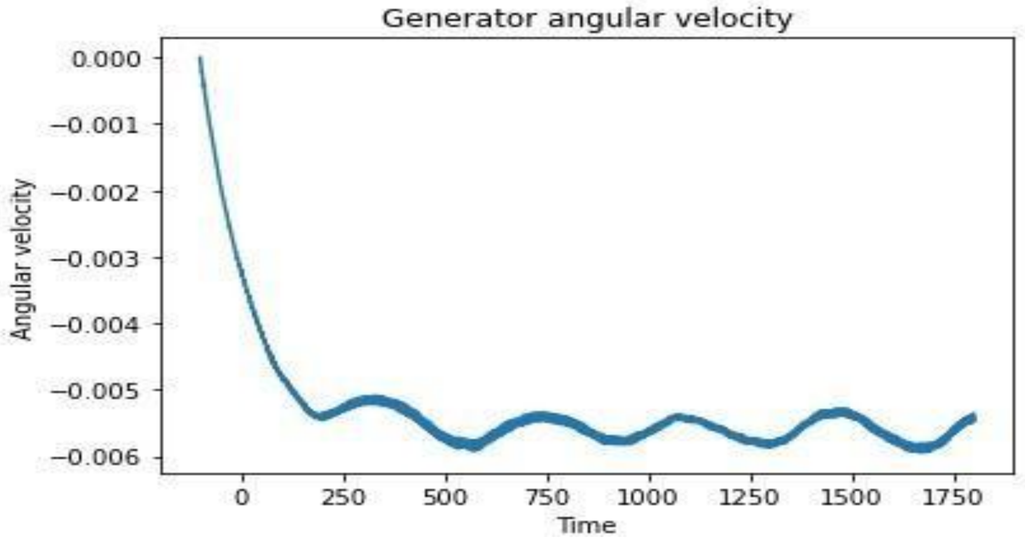
2.3 Figure 3: Rotor aero power of the 15MW wind turbine.



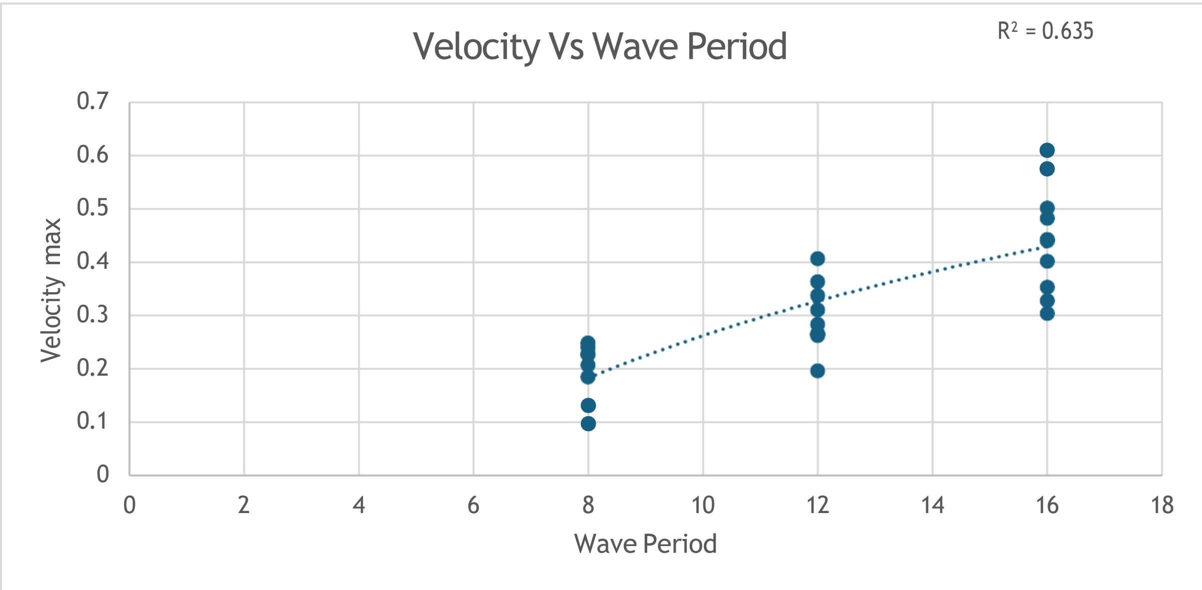
2.4 Figure 4: Rotation of Semi-Submersible platform (15MW).



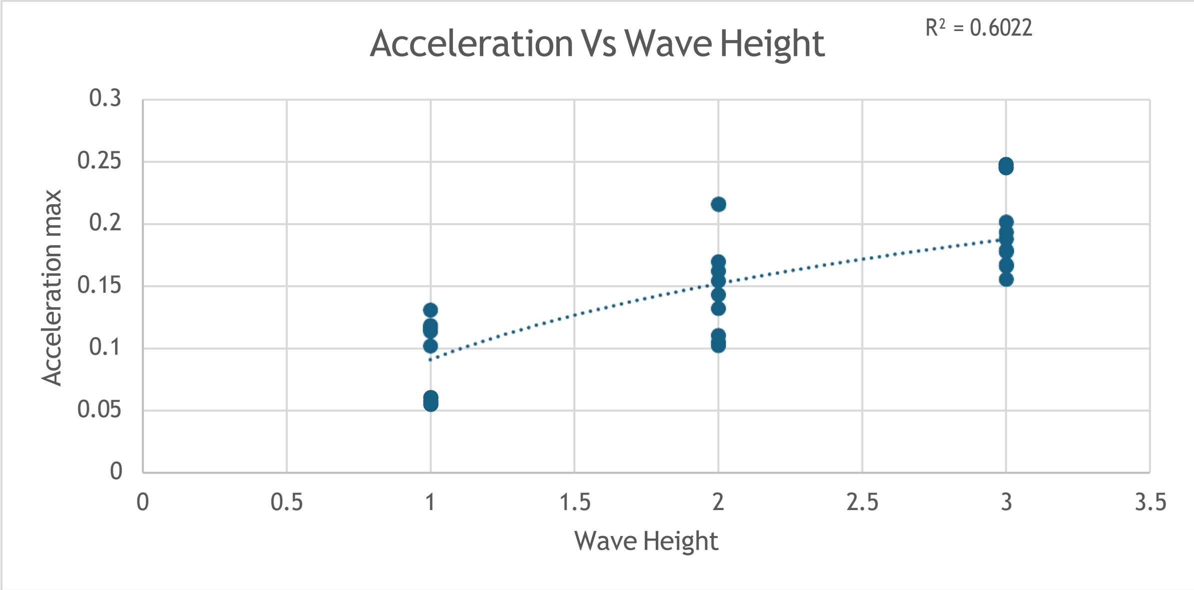
2.5 Figure 5: Rotor aero power of the 10MW wind turbine.



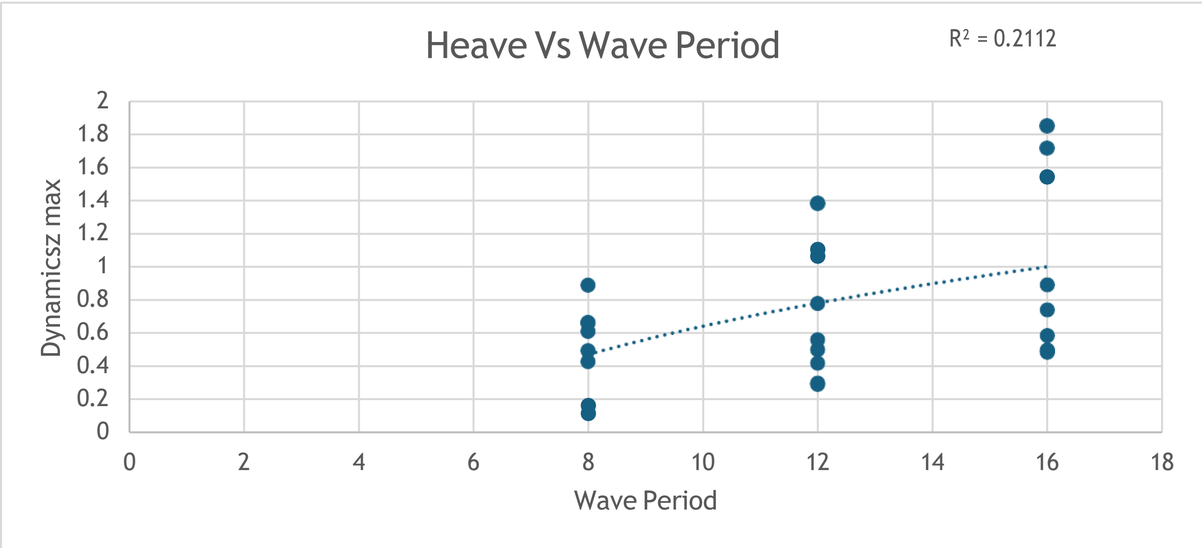
2.6 Figure 6: Generator angular velocity in 10MW wind turbine.



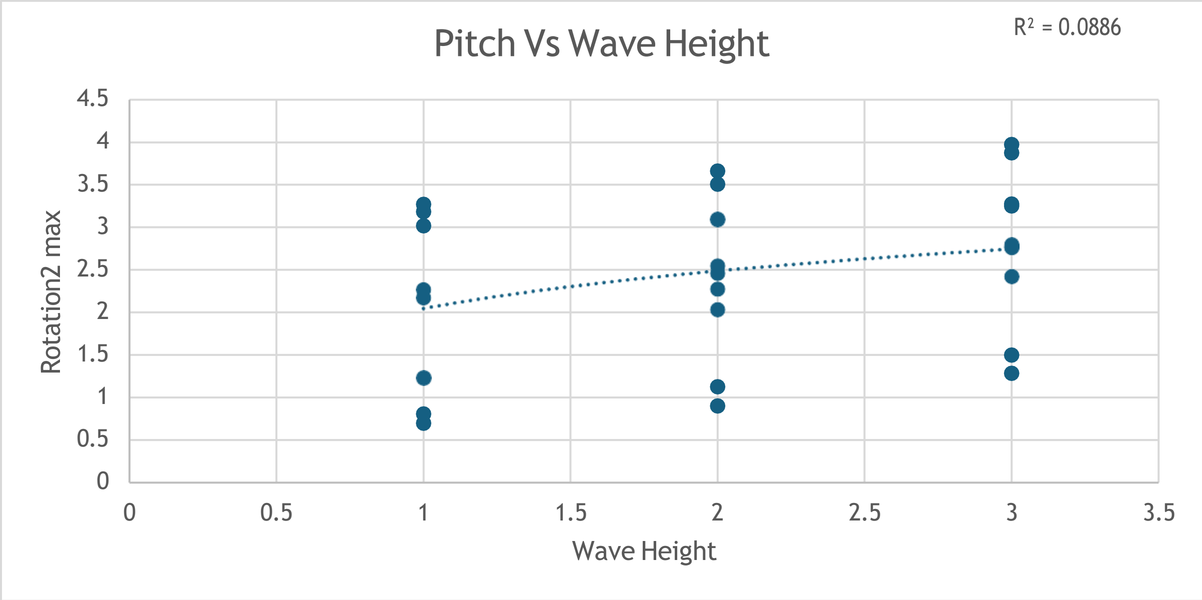
2.7 Figure 7: Significant Influence of wave period on velocity of moorings.



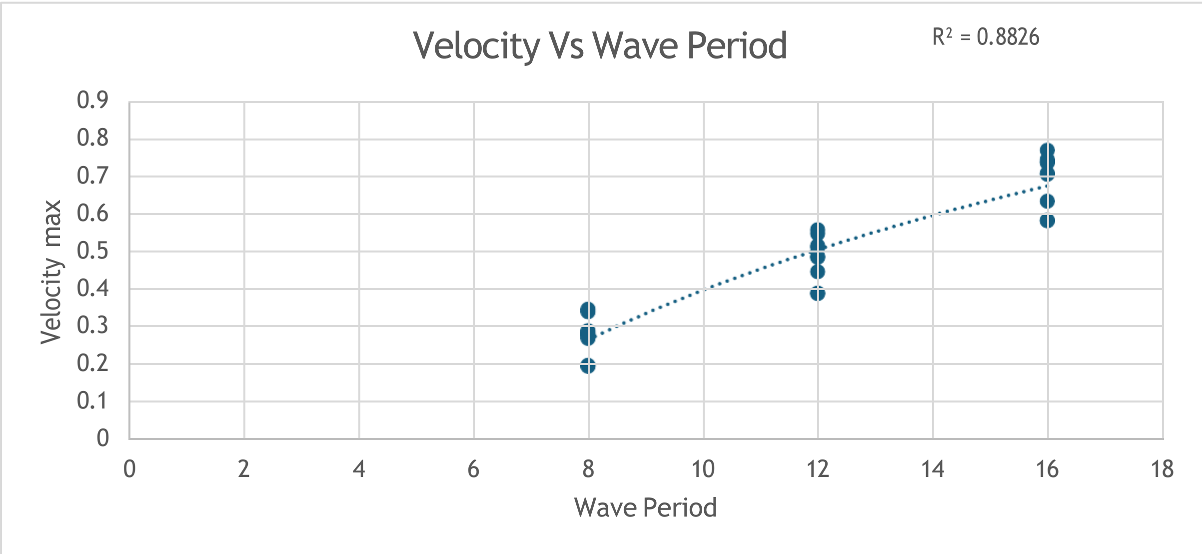
2.8 Figure 8: Significant Influence of wave height on acceleration of moorings.



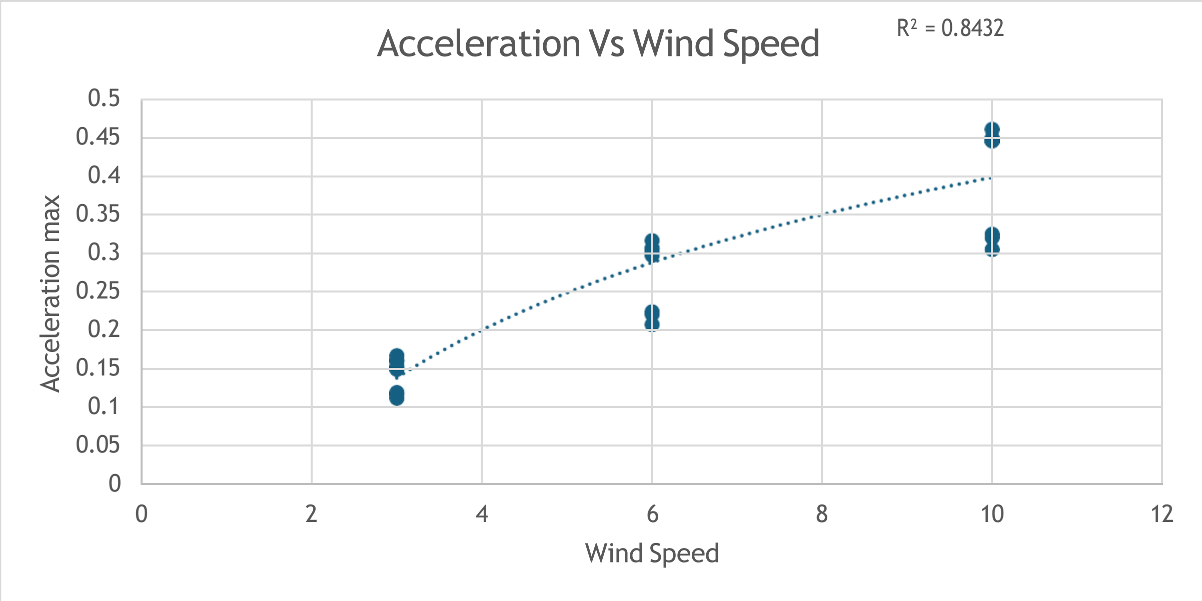
2.9 Figure 9: Influence of wave period on heave motion of buoy.



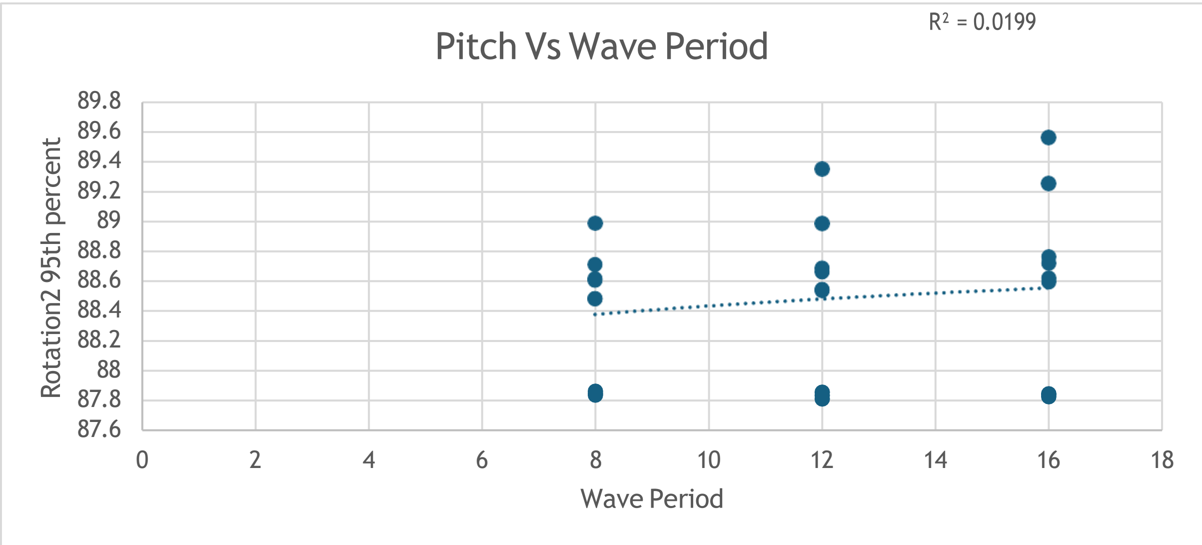
2.10 Figure 10: Influence of wave height on pitch rotation of buoy.



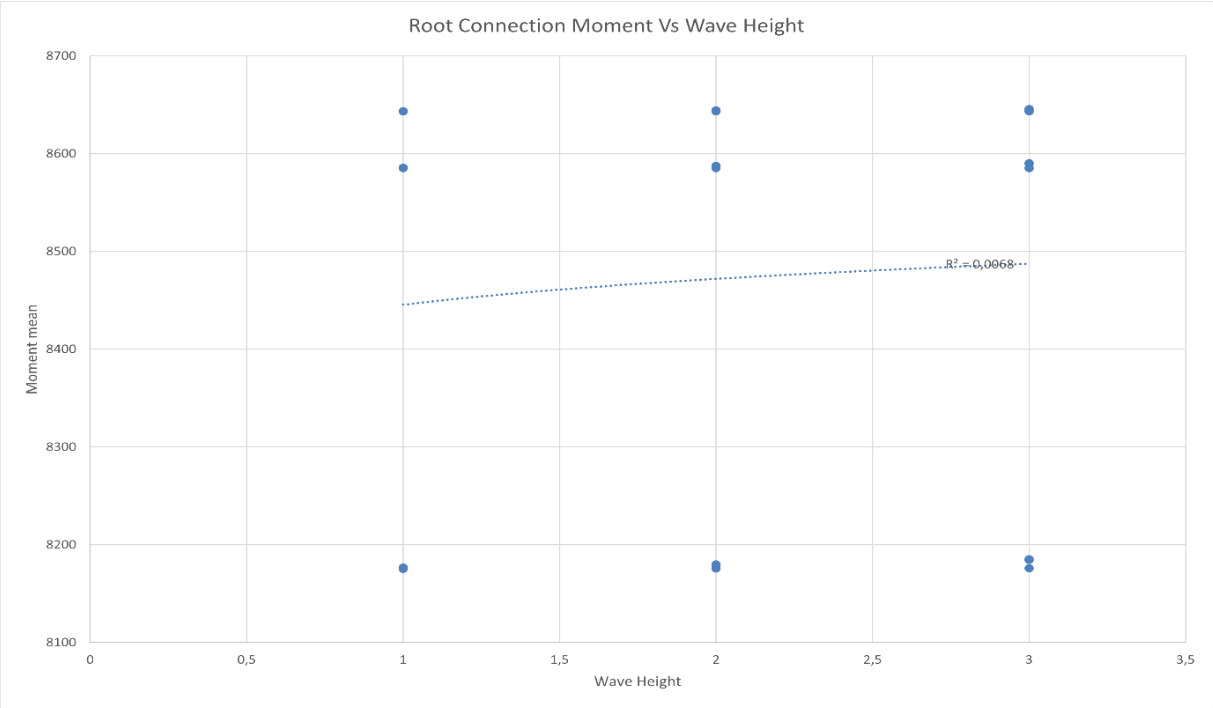
2.11 Figure 11: Significant Influence of wave period on velocity of moorings.



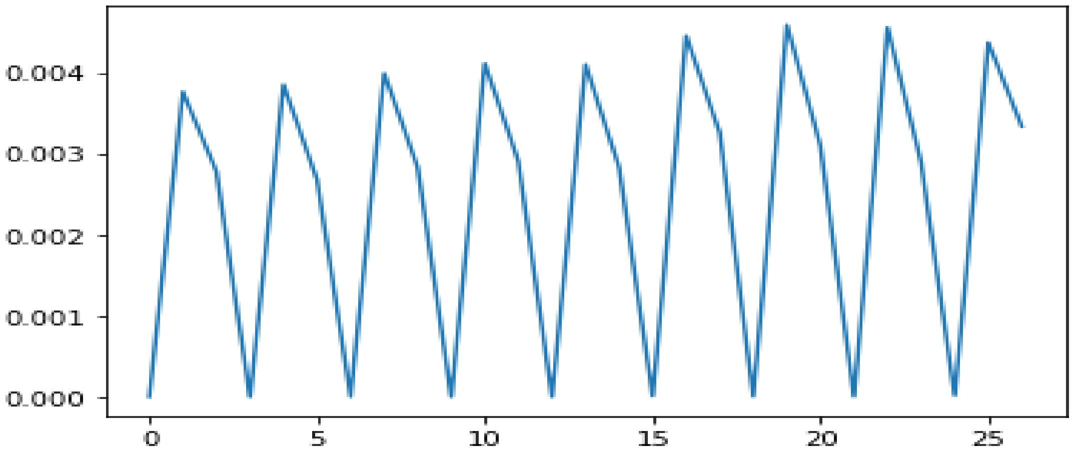
2.12 Figure 12: Significant Influence of wind speed on acceleration of moorings.



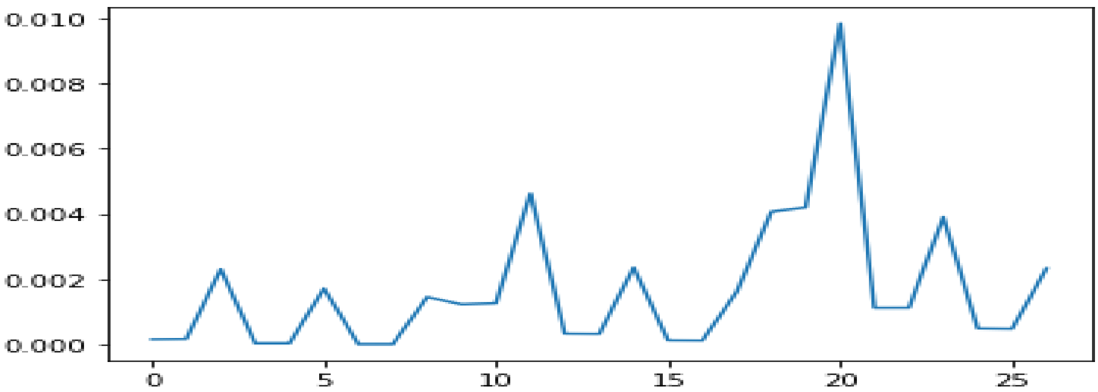
2.13 Figure 13: Influence of wave period on pitch rotation of buoy.



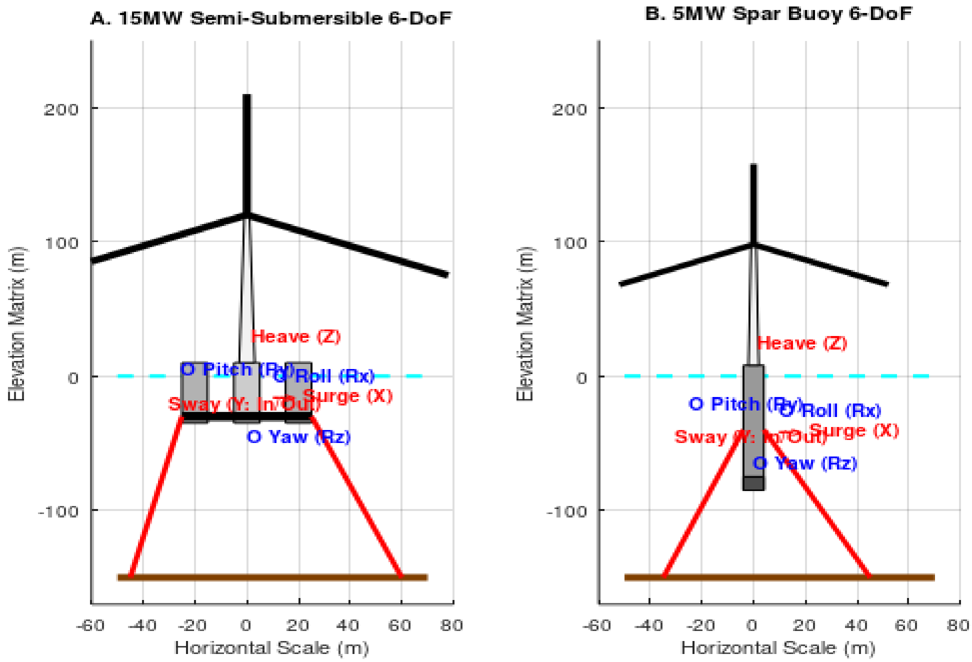
2.14 Figure 14: Influence of wave height on root connection moment of turbine.



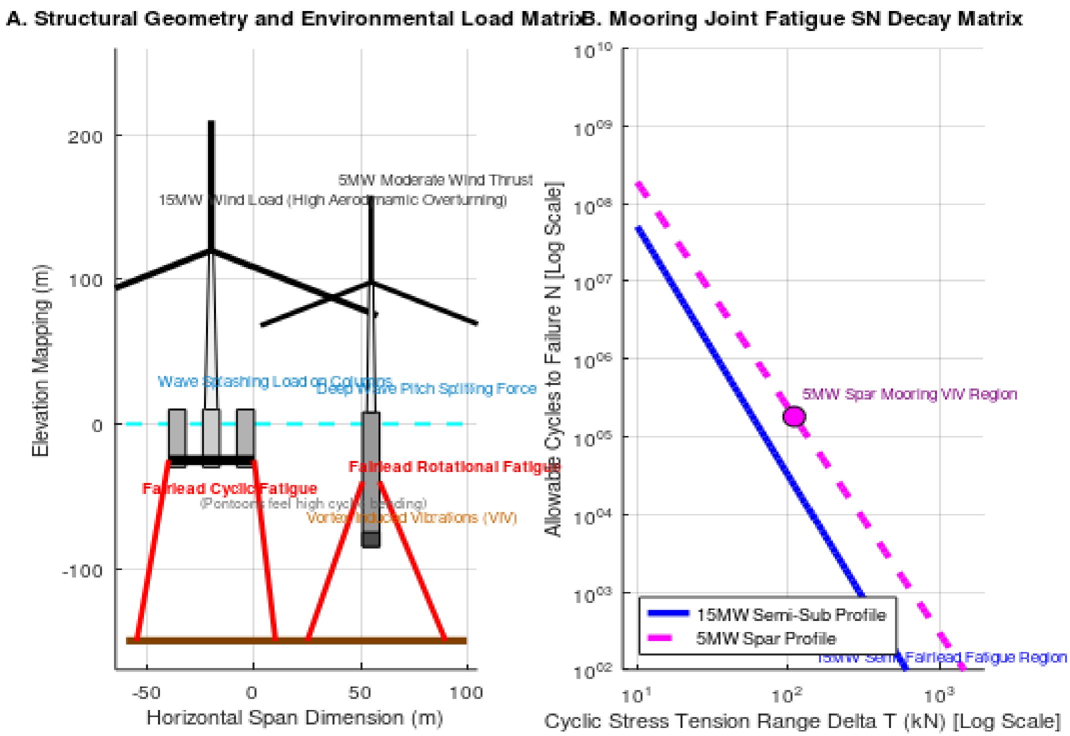
2.15 Figure 15: Fatigue damages represented by all the 27 cases for 5MW Spar type.



2.16 Figure 16: Fatigue damages represented by all the 27 cases for 15MW Semi-Submersible type.



3.1 Figure 1: 5MW Spar and 15MW Semi-Submersible type comparison based on degrees of freedom.



3.2 Figure 2: Comparison of fatigue analysis and environmental loads in 5MW Spar and 15MW Semi-Submersible type.